

Gewu Insight



User Guide

Version 0.1 (Build 673606679)

Your personal research workspace, keeping papers, evidence, and files connected over time.

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About this manual

This PDF is the fixed-layout edition of the in-app Gewu Insight User Guide. It contains the same product guidance as the searchable Help window, arranged for comfortable reading, printing, and archival use. Interface names remain in English so that the manual matches the application.

Three complementary reading paths

Operate the workspace

Parts I and III explain the interface and everyday workflows: navigation, papers and files, importing, monitors, settings, recovery, and transfer.

Understand the product

Parts II and IV explain the local-first philosophy, the visible-workspace model, SQLite's role as the source of truth, and the product's visual identity.

Inspect the technical contract

Part V documents storage, database relationships, network behavior, migration, troubleshooting, and other details needed for maintenance or development.

These paths intentionally share one manual. The user-facing behavior, the product model, and the implementation constraints can therefore be read independently without drifting into separate and contradictory documents.

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Part I.

Orientation

1-1 Welcome

Gewu Insight is a local workspace for collecting papers, preserving their files and versions, and following journals, subjects, authors, and your own publications.

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1-1.1 Why this workspace exists

Research rarely arrives as one clean collection. A paper may first appear in a feed, later acquire an arXiv version and a DOI, be downloaded under several filenames, reappear through an author's profile, and eventually become part of a project or your own publication record. Publisher pages, alerts, PDFs, folders, notes, and citation services each preserve only part of that history.

Gewu Insight is designed to keep those fragments connected. It treats discovery, identity, files, versions, organization, reading state, and provenance as parts of one continuing research process. The aim is not to accumulate the largest possible list. The aim is to preserve enough context that a useful paper remains understandable and usable months or years after it was first found.

1-1.2 Why the name 格物洞见

The name recalls the Chinese idea of 格物致知: approaching things closely enough to understand their underlying order. In this practical research setting, *gewu*—the investigation of things—means continuously examining the research object, its sources, authors, versions, files, relationships, and changing state.

Element	Role
格物 · Gewu	Stay close to the research object. Keep the paper, identifiers, versions, sources, files, and relationships inspectable instead of reducing them to an untraceable card or summary.
洞见 · Insight	Let understanding emerge from connected evidence, repeated reading, comparison, and the researcher's judgment. Insight is accumulated; it is not a conclusion the app claims to manufacture.

The Chinese name is part of the product idea, not a decorative translation of the English name.

> Keep research objects visible, evidence traceable, and insight cumulative. 让研究对象保持可见，让证据保持可追，让洞见能够积累。

1-1.3 Five design commitments

Principle	What it means
Local ownership	Your live database and managed research files remain in the local project. Online services contribute evidence; they do not own the workspace.
Visible evidence	PDFs and attachments remain real files that Finder, Preview, and other applications can inspect.
Continuity over time	A preprint, later DOI, publication version, annotations, and personal contexts can accumulate around one persistent paper object.
Explicit state and provenance	Completed, empty, partial, failed, downloaded, imported, and retried are different states. Sources and operational outcomes should remain visible.
Researcher control	Automation assists discovery and maintenance, but classification, interpretation, organization, and scientific judgment remain yours.

> **Local-first does not mean offline-only**
> The durable workspace is local, while supported discovery, metadata, profile, and download features may require network access and remain subject to source, publisher, and institutional availability. Verify uncertain matches and versions; the app does not bypass access controls.

1-1.4 Cross-platform core, platform-specific integration

Gewu Insight separates its portable Python, SQLite, and web-interface core from the native desktop layer. Paper identity, database migrations, monitor state, resource relationships, and most interface logic are designed without making macOS the definition of the product.

Platform layer	Current status
Portable core	Python, SQLite, the local application service, and the web-based workspace form the shared architectural foundation.
macOS native release	This is the current primary release and the platform on which the complete desktop workflow is developed and tested, including Finder, Preview, Quick Look, Dock, Trash, and native drag and drop.
Planned Windows release	Windows support is planned to begin as an experimental build. Explorer, the default PDF viewer, Recycle Bin, native drag and drop, packaging, and updates require their own implementation and validation.

Current distribution status
This guide does not claim that a supported Windows package is available today. Instructions that name Finder, Preview, Quick Look, the Dock, or macOS Trash describe the currently tested macOS integration. Equivalent Windows instructions should be added when the experimental Windows package exists and has been verified.

1-1.5 What this guide covers

This guide follows the work you do in the app: put papers into the Library, organize them, manage publication versions and attachments, review new literature, and keep the database and files safe. It describes the current interface rather than the history of the project.

1-1.6 What Gewu Insight is—and is not

It is	It is not only
A local research workspace that connects discovery, identity, files, versions, organization, and durable state.	An RSS reader that forgets a result after it leaves the feed.
A visible file and resource system built around persistent paper identities.	A PDF folder whose filenames must carry every piece of meaning.
A personal research memory shared by Library, Spaces, publications, and monitors.	Five unrelated tools that happen to occupy the same window.
A system that records evidence and assists repeatable work.	A black-box ranking or summary engine that replaces scientific judgment.

1-1.7 Six main tabs

- **Library** — your managed papers, folders, spaces, versions, and related files.
- **My Publications** — your publication list, citation metrics, author roles, and export tools.
- **Journal Monitor** — papers published on a selected date, organized by journal and field.
- **Subject Monitor** — topic-based discovery from supported online sources and feeds.
- **Author Monitor** — monitored researchers, profile-linked discovery, and newly found work.
- **Browser** — Google search, websites, online PDFs, hierarchical bookmarks, and local browsing history.

1-1.8 One continuous research loop

Element	Role
Acquire **Find**	Journal, Subject, and Author Monitor introduce relevant candidates.
Identify **Recognize**	DOI, arXiv, title, authors, and source evidence are connected to a shared record.
Resolve **Preserve**	Metadata, files, versions, roles, paths, and provenance are made explicit.
Remember **Understand**	Library, Spaces, My Publications, reading state, and personal organization retain meaning.

What you save, read, classify, and follow becomes context for the next cycle. The five research views share this loop; Browser supplies a separate place to explore websites.

> **Important**
> SQLite is the source of truth for Gewu Insight. Do not replace, rebuild, or edit the database by hand. Use the app to move, classify, or delete managed items.

1-1.9 Choose a path through this guide

This is one comprehensive manual with three complementary reading paths. The **operational path** explains how to use the app; the **concept and architecture path** explains why the workspace is organized this way; and the **technical reference path** documents storage, database, network, migration, and troubleshooting details. They remain together so visible behavior, product principles, and implementation constraints do not drift apart.

If you want to	Continue with
Start using the app now	Read **Part I · Orientation**, then use **Part III · User Manual** as the operational guide.
Understand the product idea and architecture	Read **Part II · Product Concepts and Architecture**, followed by **Part IV · Product Identity**.
Maintain, migrate, or inspect the system	Use **Part V · Technical and Operational Reference**, with Part II for the underlying model.
Look up one operation	Use the topic list or press `⌘F` to search this guide.
Prepare an upgrade or move to another Mac	Read Back Up or Move to Another Mac before changing the installation.

1-1.10 Open this guide again

Choose **Help > User Guide**. The guide opens in its own window and works without an internet connection.

See also: Guided Tour · Your First Session · System Architecture at a Glance · How Gewu Insight Stores Files

1-2 Guided Tour

The top chooses a work mode; the panes move from context on the left to the active object and its evidence; the bottom preserves navigation and progress without reshaping the workspace.

1-2.1 Read the interface as a sentence

Gewu Insight uses a consistent left-to-right grammar. The exact number of panes changes with the task, but their meaning remains stable: choose a research context, narrow it to an object, inspect its evidence, then act. This lets a dense workspace remain predictable without hiding the underlying files and relationships.

Element	Role
Top **Choose a mode**	Library, My Publications, one of the three discovery monitors, or Browser.
Left **Set the context**	A root, folder, date, journal, subject, author, or filter defines the working scope.
Center **Select the object**	Browse folders, paper lists, search results, or the resources belonging to one paper.
Right **Inspect and act**	Read metadata and evidence; open, download, classify, refresh, export, or reorganize.

The bottom edge is reserved for progress, breadcrumbs, and view controls so background work does not repeatedly resize the main content.

1-2.2 The Library window

Area	Purpose
1. Source tabs	Library, shortcuts, and independent Spaces.
2. Folder columns	Move from broad folders to the selected item.
3. Item column	Papers, files, and child folders in the current location.
4. Paper Files	One flat list of resources with independent type, annotation, source, and Default status.
5. Details	Metadata, identifiers, actions, citations, and folder membership.

1-2.3 Follow selection from broad to specific

A selected item does not always mean “open this file.” In a folder column, selection establishes the next level of context. In the item column, it identifies the active paper or standalone file. In Paper Files, it identifies one resource inside the active paper bundle. Paper Details then describes the shared work rather than merely repeating the selected filename.

Selection	What becomes active	What remains shared
A Library folder	Its child folders and items.	The papers keep their identities, resources, and other placements.
A paper in the item column	Its Paper Files bundle and Paper Details.	The same work can still belong to other folders, monitors, or My Publications.
A resource in Paper Files	That version or attachment for preview, classification, or file action.	The parent paper remains the active scholarly object.
A Monitor result	Its discovery evidence, report, and available actions.	Adding it to Library reuses the cataloged paper record when identity has been resolved.

1-2.4 Tabs across the top

Use the six top tabs to change workspaces. A monitor may show a calendar and filters at the left, results in the middle, and a selected-paper view at the right. Browser instead has a Bookmarks/History sidebar beside a native webpage. Progress messages appear in reserved status areas so the main layout remains stable.

On macOS, double-click only a genuinely blank part of the title bar when you want to zoom or restore the window. Double-clicking an app tab, the Library search field, or another top control is treated as interaction with that control and does not change the window size.

Area	Primary question	Typical next action
Library	What have I chosen to keep, and how is it organized?	Read, classify files, move, copy, attach, or recover.
My Publications	Which works are mine, and what authorship or citation context do they carry?	Verify roles and metrics, refresh, order, or export.
Journal Monitor	What appeared in selected journals on a given date?	Filter, inspect, download, or add to Library.
Subject Monitor	What matches a defined scientific topic across supported sources?	Review source evidence and preserve useful results.
Author Monitor	What is new in the work of a researcher I follow?	Inspect the author relationship, open the work, or add a Library placement.
Browser	What can I find on the web without leaving this workspace?	Search Google, read a website or online PDF, save a bookmark, or revisit history.

1-2.5 Watch one paper move through the workspace

Element	Role
Discovered	A Monitor records where and when the work was observed.
Cataloged	Identity evidence connects the observation to a persistent paper record.
Organized	A Library folder, Space, or My Publications adds personal context.
File-backed	Downloaded or attached resources become available in Paper Files and Finder.

These states can overlap. A cataloged Monitor result may have no local file, and a downloaded file does not have to belong to a Library folder immediately.

1-2.6 Resize a pane

1. **Point to a divider.** Move the pointer to the vertical line between two panes.
2. **Drag the divider.** Move it left or right until the useful content fits.
3. **Release.** The app uses the new working width without changing the underlying files.

> **Note**
> The right-side Paper Details area scrolls independently. A long author list or citation section does not move the Library columns.

1-2.7 Recognize stable and temporary state

Selection, keyboard focus, menus, and scroll position are temporary interface state. Paper identity, placements, resource properties, read state, provenance, and completed operations persist when saved. A breathing progress message means work is active, not that a paper or folder was created.

> **Keep the active context visible**
> Before moving, deleting, or attaching, check the focused pane and selected Paper Details. Similar names can identify different targets: a paper placement, one Paper Files resource, or a standalone file.

See also: Library Layout · Navigate with Mouse and Keyboard · Keyboard Shortcuts

1-3 Your First Session

Start with one paper, confirm its identity and context, then inspect or add a local file when the work is file-backed.

1-3.1 Choose your starting route

You do not need to prepare a perfect record before beginning. Start from the strongest evidence you already have, then let the paper object become richer over time.

You already have	Start here	What Gewu Insight does first
A local PDF or another research file	Drag it into the target Library folder, or use the folder's add command.	Preserves a visible file, creates a catalog record, and attempts to extract or resolve identity evidence.
A DOI, arXiv identifier, supported URL, or title	Choose **Add Paper** in the target folder and enter the evidence.	Creates or reuses a paper record, resolves metadata, and offers a download when a supported file is available.
A useful result in Journal, Subject, or Author Monitor	Select the result and choose **Add to Library**.	Adds a Library placement for the cataloged paper while retaining its discovery provenance.

External originals stay separate
Folder Loader and ordinary file import are designed to preserve the external source rather than silently reorganize it. The managed copy and the source path can both remain part of the provenance.

1-3.2 Add and inspect a paper

1. **Open Library.** Select an existing folder, or create one from the context menu.
2. **Add a paper.** Drag a PDF into the folder, or open the target folder, Control-click blank space in its contents, choose **Add Paper**, and enter a DOI, arXiv identifier, supported URL, or title.
3. **Select the new item.** Paper Details appears at the right. Gewu Insight attempts to identify the work. A local file is recorded only when you imported one or a supported download succeeded.
4. **Check DOI and arXiv.** Enter a corrected DOI in its field and refresh it, or edit arXiv inline and press `Return`.
5. **Inspect the available evidence.** If a local PDF exists, choose **Open in Preview**, or enter Paper Files and press `Space` for Quick Look. If no file exists yet, use **Open Website** or download a supported version when one is available.

Result. The paper now has a cataloged paper record, a stable Gewu Key, and a Library placement. It may have no local file; that cataloged-only state is valid. A file resource exists only after a file is imported, downloaded, or attached.

1-3.3 Understand what may now exist

Element	Role
Always **Catalog record**	Title, authors, identifiers, source evidence, and a stable internal paper identity.
Always **Library placement**	A relationship that makes the paper visible in the folder you selected.
When file-backed **File record**	Type, annotation state, source, stored path, fingerprints, and Default status when known.
When file-backed **Visible local file**	The actual PDF or attachment that Finder and Preview can open.

A cataloged-only paper legitimately has the first two objects and no local file. A file-backed paper also has resource metadata and real bytes. These objects cooperate, but they are not interchangeable.

1-3.4 Verify the identity before enriching it

1. **Compare the title and authors with the PDF.** A similar title is useful evidence but not sufficient proof.
2. **Check the DOI and arXiv fields.** A work can have both. Correct an inaccurate identifier before refreshing more metadata or downloading another version.
3. **Read the source and journal information.** Confirm that a publication record belongs to this work rather than a cited paper or a similarly named article.
4. **Keep uncertainty visible.** It is safer to leave a field empty than to confirm a plausible but wrong identifier.

1-3.5 Complete the first Paper Files bundle

A cataloged-only paper may have no local file, and nothing is missing merely because Paper Files is empty. Continue with these steps after you import, download, or attach a file and the paper becomes file-backed.

1. **Select the paper to show Paper Files.** Cards initially show File type and size. Click a card's size footer to show or hide Annotation and Source.
2. **Classify the available file.** Use its small type, annotation, and source controls. Do not infer these facts from the filename alone.
3. **Add related material if you have it.** Add a supplement, annotated copy, note, presentation, data file, code file, or other attachment. Import first, then classify the resource.
4. **Check the default file. Open in Preview** and Library preview use the resolved default when applicable. Selecting **Set as Default** changes preference without renaming the resource.
5. **Reveal one file in Finder.** Choose **Show in Finder** and confirm the real managed file exists where the database expects it.

Principle	What it means
Cataloged and organized	The shared paper record and Library placement are durable even when no local file has been added.
File-backed	After a file is added, its independent file properties and physical bytes can be inspected separately from paper identity and folder context.

1-3.6 Optional: verify shared identity

Copy the paper to a second Library folder, then select it there. The title, Gewu Key, Paper Files, and metadata should remain the same because the second folder contains another placement for the shared paper—not an unrelated duplicate. Delete only the extra placement when you are finished with the check.

1-3.7 Next useful actions

- Use **Download** when a better publication version is available.
- Use **Paper Files** to classify imported PDFs or add notes, data, slides, and supplements.
- Add the paper to another folder without creating a second independent work record.
- Back up the project before a large import or migration.

1-3.8 Finish with a safe baseline

Your first session is complete when you can answer two questions for every paper: What is this work, and why is it in this folder? If the paper is file-backed, answer two more: Which file is the default, and where are the real bytes? Paper Details, Paper Files, and Finder should answer the applicable questions without contradicting one another.

Principle	What it means
Identity checked	The title, authors, DOI, and arXiv evidence do not point to different works.
Placement understood	The Library folder expresses personal organization, not a second scholarly identity.
Resource classified when present	If the paper is file-backed, its files can have independent type, annotation, and source descriptions.
File visible when present	If the paper is file-backed, the expected managed file opens from the app and can be revealed in Finder.

Create the backup habit early
Before a major import, upgrade, or migration, quit Gewu Insight and back up both the SQLite database and the managed file roots. A database-only copy lacks some bytes; a files-only copy lacks the relationships and state that make the workspace intelligible.

See also: Add Papers and Files · Use Paper Details · Understand Paper Files · Back Up or Move to Another Mac

Part II.

Product Concepts and Architecture

2-1 System Architecture at a Glance

Gewu Insight is a local research system that coordinates structured state, visible files, and a personal workspace. It is not simply a web page placed in front of a folder.

2-1.1 How the parts cooperate

Element	Role
Native desktop shell	Starts the app, owns Help and Browser, and connects Preview, Finder, and native file actions to the portable core.
Personal Research Workspace	Library, Spaces, My Publications, and the three Monitor views you see and operate.
External research sources	Publisher pages, feeds, metadata services, scholarly profiles, and preprint services.
Local application service	Interprets your actions, resolves identities, schedules network work, and coordinates database and file changes.
SQLite state core	Owns identity, relationships, resource descriptions, preferences, provenance, and durable workflow state.
Visible Research Files	PDFs, versions, supplements, notes, data, and other files that remain inspectable in the native file manager and document applications.

The interface sends an intention to the local service. The service validates it and coordinates SQLite with the relevant files; the UI does not directly treat a Finder folder as the database.

The desktop window presents a local interface served from the project on the current computer. In today's primary release, a macOS shell supplies Finder, Preview, Quick Look, Dock, and native drag-and-drop integration. The Python, SQLite, and web-interface core is designed to support other desktop shells without changing the meaning of the research data. Network sources contribute evidence and downloadable material, but the durable workspace stays local.

2-1.2 What each persistent layer answers

Browser is a separate web surface. Its native WebKit views load websites directly, not through the Python service. The local service stores bookmarks and history in SQLite; WebKit manages a separate persistent website profile. External pages do not receive the Library's local API bridge. Visiting a page, saving a bookmark, or downloading through Browser does not create a paper or assign a Gewu Key; import the paper explicitly.

Layer	Questions it answers	What it does not decide alone
SQLite	What is this paper? Where is it referenced? Which resources belong to it? What happened in the last operation?	It does not contain the bytes of your PDFs and attachments.
Filesystem	Which real files can the native file manager, PDF reader, or another application open?	A filename or folder position alone does not define paper identity, version priority, or monitor membership.
Personal Research Workspace	How should the shared state be presented for discovery, reading, organization, and action?	It is a projection of saved state, not a separate copy of the collection.

> System relationship
> Persistent research workspace = structured SQLite state + visible research files, coordinated by the local application.

2-1.3 Why this separation matters

A paper can appear in a Monitor, the Library, a Space, and My Publications without becoming four unrelated papers. It can also acquire a DOI after first arriving as an arXiv preprint, or acquire a publication PDF while retaining earlier versions. The database preserves continuity across those changes; visible files preserve user ownership and inspectability.

> **Terminology**
> This guide uses **Personal Research Workspace** for the on-screen projection and **Visible Research Files** for assets visible to the native filesystem. These are explanatory terms, not menu names. Together they form the visible workspace, but they serve different roles.

See also: The Persistent Paper Object · Why SQLite Is the Core · Visible Research Files

2-2 The Persistent Paper Object

Gewu Insight treats a scholarly work as an object that persists through discovery, publication, file changes, and personal organization—not as a temporary search result or a single PDF.

2-2.1 One identity can accumulate evidence

Element	Role
Identity evidence	DOI, arXiv identifier, title, authors, journal, dates, and source identifiers.
Resources	Publication PDFs, preprints, accepted manuscripts, supplements, annotations, and attachments.
Contexts	Library folders, Spaces, My Publications, and Monitor relationships.
Durable state	Read status, metadata quality, provenance, download state, retries, and timestamps.
One shared paper identity · one stable Gewu Key	The internal identity stays stable when an external identifier, filename, folder location, or preferred PDF changes.

Identifiers help resolve and enrich a paper, while the Gewu Key provides continuity inside Gewu Insight.

The central record uses an internal paper identity. A DOI or arXiv identifier is important evidence attached to that identity, but neither is a safe universal primary key: an early preprint may have no DOI, an imported PDF may initially have no reliable identifier, and an incorrect identifier may later be corrected.

2-2.2 Do not collapse four different things

Object	Meaning	Example
Paper record	The shared scholarly identity and normalized metadata.	The work that later connects arXiv and a journal DOI.
Placement or relation	Why and where the paper appears in your workspace.	A Library folder entry, a shortcut, a My Publications role, or an Author Monitor association.
Resource record	A described file or retrievable version belonging to the paper.	Preprint v2, published version, supplementary material, peer review file, or data file.
Physical file	The bytes stored at a real path on disk.	A PDF under data/Library or an associated attachment.

> System relationship
> Paper record ≠ workspace placement ≠ resource record ≠ physical file.

2-2.3 What happens when the same paper is found again

Journal, Subject, and Author Monitor can encounter the same work from different directions. Identity resolution is designed to reuse one shared paper record and add source or context relationships. Copying a paper to another Library folder likewise creates another placement rather than a second independent identity. Its metadata, Gewu Key, versions, and attachments remain shared.

When the available evidence is insufficient or contradictory, separate candidates can remain until their identifiers or metadata are corrected. A major DOI, title, author, or venue contradiction is quarantined as an identity conflict instead of overwriting the saved paper or forcing a merge. The shared-object model is the resolution target; it does not justify merging uncertain works merely because their titles look similar.

A network observation can be transient while acquisition is running. Once a result is cataloged as a shared paper record, it has a Gewu Key even if a particular Monitor view does not display that key. A Monitor result may still have no local Paper Files because no file has been downloaded or attached; adding it to Library creates a visible placement rather than inventing a new scholarly identity.

Gewu Insight also uses the same catalog and resource infrastructure for an imported standalone file whose scholarly identity is not yet known. In that case, the catalog record preserves the file and its relationships while later identification can enrich it.

2-2.4 Identity corrections should preserve continuity

Editing a DOI or arXiv identifier changes evidence on the existing paper. It should not silently make an unrelated second paper, rename every file, or discard your folder placements and read history. When evidence conflicts, inspect the title, authors, venue, and files before confirming the correction.

> **Important**
> Similar titles are not proof of identity, and matching filenames are not proof that two resources are the same. Use identifiers, metadata, file fingerprints, and provenance together.

See also: Edit DOI and arXiv Identifiers · Understand Paper Files · The Personal Research Workspace

2-3 Why SQLite Is the Core

SQLite is the source of truth for Gewu Insight because it preserves the identity, relationships, resource registry, and operational memory that neither a filename nor an exported report can express.

2-3.1 Four families of structured state

Element	Role
SQLite state authority	A local transactional database stored with the project.
Identity and metadata	Shared papers, external identifiers, normalized title and authors, dates, fields, and Gewu Key.
Relationships and context	Library placements, folder tree, Spaces, My Publications, and Monitor associations.
Resource registry	File types, paths, source evidence, hashes, annotation state, and default preferences.
Operational memory	Fetch runs, import batches, source tasks, reads, settings, failures, and retries; also Browser bookmarks and history.

Table names and schema versions can evolve; these four responsibilities are the stable user-level model.

The database does not store the PDF bytes themselves. It stores enough structured information to understand the files: which paper they belong to, what file type they represent, where they are, where they came from, whether they are annotated, and which one should be used by default.

2-3.2 Why a folder scan is not an equivalent database

Finder can show that `paper.pdf` exists. It cannot reliably say whether it is a published version or a preprint, whether it is annotated, where it came from, whether the same work is also in My Publications, or which file is the manual default. Those facts live in SQLite.

2-3.3 Transactions, relationships, and WAL

Database mechanism	User-facing purpose
Transactions	Keep a multi-step state change together so a partly applied edit is less likely to become the visible truth.
Foreign-key relationships	Connect a resource or placement to the correct shared paper instead of relying on a coincidentally similar filename.
WAL (write-ahead logging)	Lets foreground reading and background updates coexist more safely than repeated full-file rewrites.
Schema migration	Adds new structures and backfills required state during an upgrade rather than asking the user to rebuild the database.

WAL improves normal concurrency; it is not a substitute for a coherent backup. Close Gewu Insight before copying its live database, and back up the managed files with it.

2-3.4 Durable preferences and shipped defaults

Journal subscriptions are durable SQLite state as well. The database preserves which journals are enabled, their identifiers and source settings, and their display order. The journal catalog shipped with a newer app version is a set of defaults, not an instruction to overwrite your personal list.

On the first upgrade that introduces durable subscriptions, Gewu Insight seeds the database from the existing journal configuration and history. After that one-time seed, adding, disabling, restoring, or reordering a journal changes the database record. Replacing source code or rebuilding the app does not silently re-enable a journal you removed or discard journals you added.

> **Cross-layer changes**
> SQLite transactions protect database state. A file move and its database update cannot be one native SQLite transaction, so the local service coordinates the steps and uses validation or compensating recovery when a filesystem operation fails.

2-3.5 Exports and provenance

JSON, HTML, Markdown, Word, LaTeX, and PDF outputs are derived artifacts. They can be regenerated, shared, or inspected, but editing them does not alter the source records. Raw source evidence may also be retained inside SQLite for provenance; this does not turn an external JSON export into a second source of truth.

2-3.6 Paper-level Monitor provenance

Normalized paper-source observations are the authority for how a canonical paper was encountered. Each observation identifies a source kind, provider name, and stable source key, and keeps first-seen and last-seen times. The same paper and effective source update one observation instead of increasing the provenance count indefinitely.

Journal Monitor also connects each observed paper to the relevant fetch run. The run supplies the journal, requested date window, status, and timing; the paper-level observation records that the reporting source saw this particular work. Subject and Author Monitor relationships can point to the same canonical paper, so three discovery routes add context without manufacturing three scholarly identities.

A bibliographic metadata refresh can add a newly proven source, but it cannot erase an earlier Journal, Subject, or Author observation. Display counts and the older raw-source list are compatibility projections of the normalized observations rather than an independent truth. When two paper identities are safely merged, their source observations move to the surviving canonical paper and duplicate observations converge.

> **Discovery is not storage**
> A provenance relationship does not by itself create a Library placement or a local PDF. It records how the work entered view. Organization and file acquisition remain separate, explicit actions.

2-3.7 What the app does during an upgrade

On startup, the storage layer reads the database schema version, checks for required tables and fields, applies additive migrations in order, ensures required roots exist, and backfills stable keys or resource records where older data needs them. A normal migration preserves paper identities, folder placements, monitored-journal choices, fetch provenance, and file paths. This is why a project upgrade should preserve the existing database and files rather than replace them with an empty database.

> **Do not edit the live database manually**
> A seemingly small manual change can violate a relationship that is not visible in one table. Use the app for normal work and make a closed, complete backup before any diagnostic intervention.

See also: Back Up or Move to Another Mac · Local Data and Privacy · Storage Reference

2-4 Visible Research Files

Your PDFs and attachments remain real local files. Gewu Insight adds a resource model around them so several versions and related materials can belong to one paper without hiding them inside an opaque cloud library.

2-4.1 One paper, one file bundle

Element	Role
Persistent Paper Object	Owns the resource relationships and decides which usable file is the default.
Default file	The automatically ranked or explicitly preferred valid file used by Open in Preview and Library preview.
Associated Files	Earlier versions, supplements, annotated copies, presentations, notes, data, code, images, and other attachments.
Paper Files	Flat peer records with independent type, annotation state, and source.
Supplements	Publisher or author-provided supporting files.
Files and figures	User material such as notes, slides, data, code, and extracted or attached images.

The Paper Files pane presents this as a compact bundle rather than another Finder hierarchy.

2-4.2 Managed and linked resources

Storage mode	Meaning	Safe expectation
Managed	The active stored path is inside the Gewu Insight project data area.	Move, classify, replace, or delete it through the app so paths and relationships change together.
Linked	The active stored path remains outside the managed project data area.	Availability depends on that external path, and the app should not silently reorganize it.

A resource can be managed while retaining a separate external source path as provenance. The existence of that source path does not by itself make the managed copy a linked resource, and the external original remains protected.

Folder Loader is a migration-oriented workflow. It imports a visible collection as a collection instead of asking you to classify every incoming file as an attachment before identity resolution. The original source path and the managed result remain conceptually distinct.

2-4.3 How the default file is resolved

An explicit manual default takes precedence while it still exists and is not classified as Invalid file. Otherwise, usable files are ranked as Corrected published version, Published version, Accepted manuscript, then Preprint. Secondary and unknown types remain visible peers but never become automatic defaults.

Supplementary material, **Peer review file**, **Other**, **Unknown**, and **Invalid file** do not become the automatic default. Any valid resource can still be selected manually; **Invalid file** is never eligible and remains visible only for audit and repair.

> **Published files**
> The user-facing type is **Published version**. A later formal correction can be classified as **Corrected published version**; the correction notice itself remains a separate peer.

2-4.4 Fingerprints and annotations

When a PDF is imported, attached, or downloaded, its exact SHA-256, content-oriented fingerprint, annotation count, and annotated state are computed as the resource is registered. They do not depend on a later refresh. Exact SHA-256 answers "are these bytes identical?" Adding a highlight or note normally changes those bytes. The content fingerprint is designed to recognize the same underlying document while discounting annotation-only changes, allowing clean and annotated copies to be related without treating them as blindly interchangeable.

Gewu Insight counts user-created comments and markup—such as highlights, notes, drawings, stamps, and form edits—as annotations. Structural PDF elements such as publisher hyperlinks, internal navigation links, and ordinary page actions do not make a freshly downloaded publisher or arXiv PDF **Annotated**.

An exact duplicate can be reused instead of stored twice. A clean PDF and an annotated copy may have different exact hashes but the same content fingerprint; both can be retained when the annotation state is meaningful.

2-4.5 Why Finder and the UI do not look identical

The filesystem must expose practical paths and folders; the UI must expose the meaning of one paper bundle. A Library paper can therefore retain a compatibility primary file in the visible topic folder while related files live under a mirrored Associated Files path. Paper Files deliberately presents all resource records as flat peers instead of mirroring those physical directories.

> **Important**
> Do not manually move or rename managed resources in Finder. The bytes may still exist, but the saved path and live workspace can disagree. Use an app-supported move, repair, or restore; do not assume a restart will rediscover an arbitrary manual move.

See also: Understand Paper Files · Choose File Types and the Default File · How Gewu Insight Stores Files

2-5 The Personal Research Workspace

The five research views—Library, My Publications, Journal Monitor, Subject Monitor, and Author Monitor—share research objects. Browser is the sixth main tab, with its own website session. Moving between research views does not reset a paper's identity.

Element	Role
Journal Monitor	Discovers work by publication venue and date.
Subject Monitor	Discovers work through topic definitions and supported feeds or searches.
Author Monitor	Discovers work through a researcher relationship.
One shared paper identity	Metadata, identifiers, resources, provenance, and state accumulate here.
Library and shortcuts	A durable managed collection with logical placements and fast entry points.
Spaces	Independent organizational roots for projects, courses, or collaborations.
My Publications	Your authorship, role, metrics, ordering, and export context on the same paper.

The same paper can participate in several contexts. Each context adds meaning; it does not require an unrelated duplicate record.

2-5.1 Views are relational projections

A Library folder answers "where do I organize this?" My Publications answers "how does this relate to my authorship?" Author Monitor answers "how does this relate to this researcher?" These are different relationships to the same possible work. The interface queries those relationships and projects the relevant controls, status, and files for the current task.

2-5.2 Library placements and shortcuts

A paper can have more than one Library placement. Copying it to another folder reuses the shared identity and resources. A shortcut tab is another entry point to a Library location, not another paper collection. A Space is an independent organizational root, but papers placed there can still reuse the same identity model.

2-5.3 Discovery does not require immediate file management

A Monitor result may remain a discovery relationship with metadata and status but no local Paper Files. A cataloged work already has a Gewu Key, which Author Monitor also shows. Downloading or attaching a file creates a resource relationship; adding the paper to Library creates a managed placement. Neither action should manufacture a second work merely because the user entered from another tab.

2-5.4 Why the same action can have different scope

Removing an author from Author Monitor removes that monitored-author relationship. Removing a shortcut removes an entry point. Moving a Library placement changes organization. Deleting a managed resource changes the paper's file bundle. The selected view supplies scope, while SQLite preserves what remains shared.

> **A useful mental model**
> The UI shows a personal map of a shared object graph. It is "visible" because you can navigate and act on it, but SQLite—not pixels on the screen—preserves the graph.

See also: Folders, Shortcuts, and Spaces · My Publications · State, Scope, and Safe Change

2-6 From Discovery to Durable Memory

Gewu Insight connects monitoring, identity resolution, version handling, and personal organization into one repeatable research loop.

Element	Role
1. Acquire	Journal, Subject, Author, Folder Loader, or a direct identifier introduces a candidate.
2. Identify	DOI, arXiv, title, authors, source evidence, and existing records are reconciled.
3. Resolve	Metadata, file types, annotation states, sources, paths, hashes, and a default preference are established.
4. Remember	Placements, authorship, reading state, provenance, completion, and failures persist.

The loop is continuous because a work can acquire better evidence, a new publication version, or a new personal context later.

2-6.1 A paper can change without losing its history

1. **Acquire an early work.** Subject Monitor finds an arXiv preprint. Its source and discovery context are stored.
2. **Identify the work.** The title, authors, and arXiv identifier connect it to one shared paper identity.
3. **Resolve its first resource.** A preprint PDF becomes the automatic default because no higher-priority file is available.
4. **Remember your context.** You add it to a Library folder, read it, and attach notes or data.
5. **Acquire new evidence later.** Journal Monitor or a metadata refresh finds the published DOI and a published file.
6. **Resolve without erasing.** The published file becomes the automatic default unless a manual choice is sticky; the preprint and its provenance remain peer records.

2-6.2 Operational state is part of scientific memory

A result count alone is ambiguous. The app distinguishes work that was never attempted from work that completed with no results, partially completed, failed, or entered a cooldown before retry. It also remembers imports, downloads, metadata refreshes, source tasks, and read state so the next action can be informed rather than blindly repeated.

State	Meaning	Reason to preserve it
Empty and complete	The requested source or date was checked and returned no matching records.	Prevents “zero” from being mistaken for “not run.”
Partial	Some sources completed while another did not.	Allows a targeted retry instead of discarding useful completed work.
Failed	An operation produced a specific error.	Makes the failure visible and avoids pretending the result is authoritative.
Cooldown or retry scheduled	A remote service asked the app to wait or a transient error needs backoff.	Reduces repeated traffic and protects normal foreground use.

2-6.3 Background work should not seize the foreground

Fetches, metadata refreshes, and imports report progress in reserved status areas and save their state. You should still be able to browse already stored material. Where callers share an online service, service-specific pacing, locks, cooldowns, and retry policies coordinate their requests. These controls reduce unnecessary traffic; they cannot guarantee that a publisher will never rate-limit or block a network address.

> **The loop is user-directed**
> Gewu Insight preserves the consequences of what you monitor, save, read, ignore, classify, and publish. It does not silently replace your research judgment with an invisible global keyword filter.

See also: Journal Monitor · Subject Monitor · Author Monitor

2-7 State, Scope, and Safe Change

A safe local research workspace changes only the object and relationship you intended while preserving shared identity, resources, provenance, and recoverable state.

2-7.1 Eight invariants behind normal operations

Principle	What it means
One shared identity	The same scholarly work should not fragment merely because several sources or folders found it.
Many contexts, not blind copies	Folders, Spaces, monitors, and My Publications add relationships around the shared paper.
Deletion has scope	Removing one relationship should not destroy unrelated relationships or shared resources.
State survives restart	Saved reading and workflow state, retries, failures, versions, and provenance are durable.
Repeated operations converge	Refreshing, importing, or downloading the same evidence should not create endless duplicates.
Good metadata is protected	A weak or empty response should not silently erase stronger saved evidence.
Database and files agree	Saved resource paths should point to the intended real files after an app-managed move.
Failure remains inspectable	Deletion, import, move, and network failure should preserve enough state to recover or retry.

2-7.2 Read an action by its scope

Action	Intended scope	What should remain
Remove an Author Monitor profile	The monitored-author relation and its profile context.	Shared papers, Library placements, My Publications, and managed files.
Remove a shortcut	That fast entry point.	The underlying Library folder and its papers.
Copy a paper to another folder	Add another placement.	One shared paper identity and resource bundle.
Move a paper in the app	Change its managed placement and corresponding stored paths.	Gewu Key, metadata, versions, and associations.
Make another file default	Change the reading preference without physical promotion.	The file's original name, location, and all other saved resources.
Delete an associated file	That resource relationship and its managed file, subject to confirmation.	The paper identity and unrelated resources.

2-7.3 Recovery is part of the design

Recycle keeps ordinary removals recoverable. Import batches and operation status provide evidence about what ran. Resource paths, hashes, and provenance help diagnose missing or duplicate files. The app can migrate older storage structure forward, but migration is not a reason to discard the previous database or original files.

2-7.4 Back up one coherent research asset

A reliable backup contains the SQLite database and the managed file roots from the same closed application state. Copying only the database preserves descriptions without every artifact; copying only folders preserves bytes without all identities, relationships, classifications, and operation history. A portable desktop copy is useful for sharing visible folder material, but it is not automatically a complete live-workspace backup.

> **Before a major update or migration**
> Quit Gewu Insight, preserve the entire project data set, verify the backup can be read, then update the software in place. Do not create a fresh empty database and attempt to reconstruct the workspace from export files.

See also: Delete and Recover Items · Back Up or Move to Another Mac · Troubleshooting

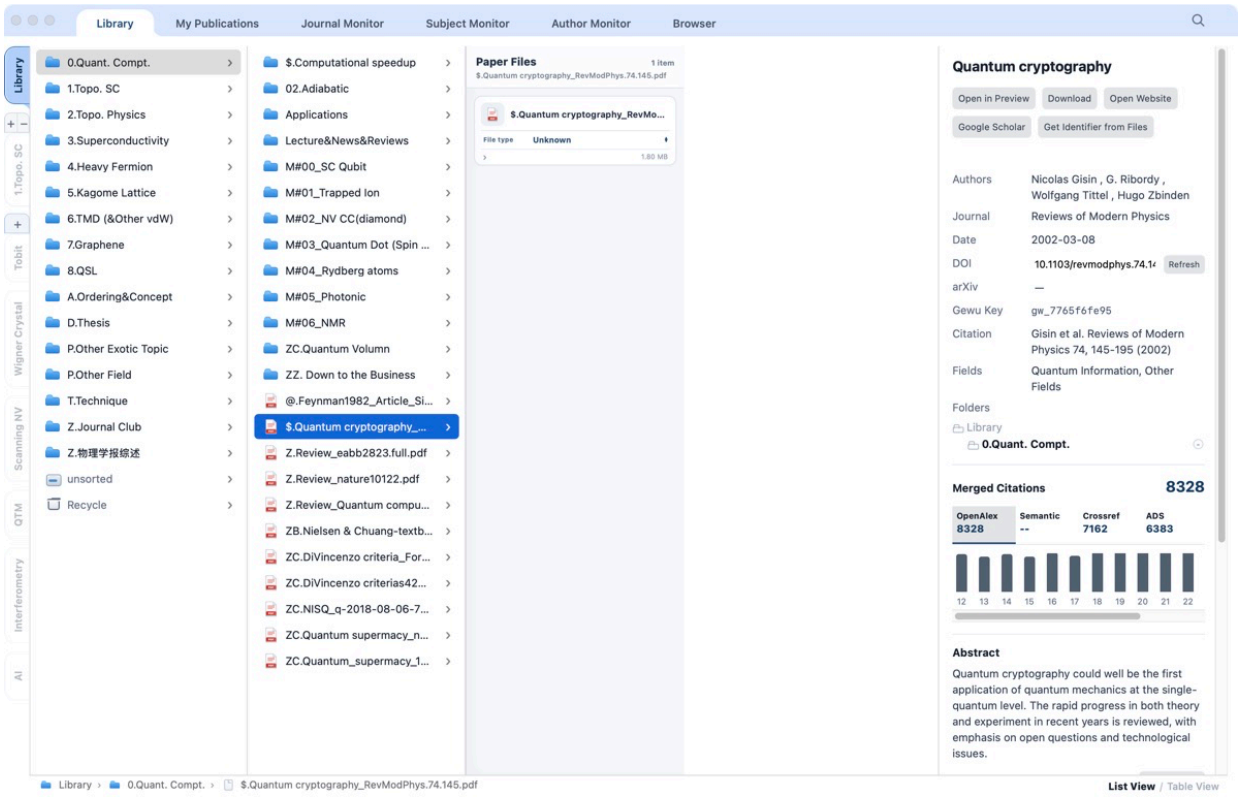
Part III.

User Manual

3-1 Workspace Navigation

3-1.1 Library Layout

The Library uses a Finder-style column view, but the database—not the visible folder alone—defines the paper and its relationships.



Library and Paper Files. The selected paper remains in the managed folder hierarchy while its related files, resolved metadata, and citation evidence stay connected in the adjacent views.

Read the columns from left to right

1. **Source:** the main Library, a shortcut, or a Space.
2. **Folders:** each selection opens its children in the next column.
3. **Items:** the papers and other files in the selected folder.
4. **Paper Files:** shown only when a paper or managed file can expose related files.
5. **Paper Details:** metadata and actions for the selected work.

Use the breadcrumb

The breadcrumb at the bottom shows the active path. Select a segment to return to that location. The **List View / Table View** control at the lower right changes the representation of items without moving them.

Folder Details

When you select a folder, the right side shows its name, type, and the number of child folders and items. This is informational; it is not another folder level.

See also: Folders, Shortcuts, and Spaces · Search, List View, and Table View · Understand Paper Files

3-1.2 Navigate with Mouse and Keyboard

Library and Space columns form one keyboard path with Paper Files. Focus moves into Paper Files with the Right Arrow and returns to the selected paper with the Left Arrow.

Move through folders and papers

- Press ↑ or ↓ to change the selection in the focused column.
- Press → on a folder to enter it.
- Press ← to return to its parent.
- Double-click a Library paper row to open its Default file: PDFs open in macOS Preview; other formats use their macOS application. An unclassified legacy import falls back to its available current file.

Move between the source rail and Library columns

1. **Return to the leftmost Library column.** Press ← once more to move focus to the vertical source rail.
2. **Press ↑ or ↓ on the rail.** Move through Library, shortcuts, and Spaces. The highlighted source opens immediately; you do not need to press Return.
3. **Press →.** Focus enters the first visible Library column for that source. Its active row changes from the quieter background selection to the stronger blue content selection.

While the source rail owns focus, the current row in the adjacent column remains visible in a quieter blue so you do not lose your place. The rail keeps focus while you move between sources and releases it only after you press →. Once you are navigating Library content, pressing ↑ at the first row stays in the Library instead of unexpectedly moving to the app tabs across the top.

Enter and leave Paper Files

1. **Select a paper in the main item column.** Paper Files appears to its right. No Paper Files card is highlighted yet.
2. **Press →.** Focus enters Paper Files and the active card gains a light-blue selection. The main paper remains visible with a quieter selection color.
3. **Press ↑ or ↓.** Move through the flat Paper Files list.
4. **Press ←.** Focus returns to the main paper. Paper Files no longer shows an active card.

The same focus rules apply in the main Library, a shortcut, and every Space. Paper Files is a database-backed view of the selected work rather than a physical child folder, but the keyboard transition is intentionally continuous: → enters the bundle, ← returns to the same paper, and the next ↑ or ↓ moves in the main item column.

Preview while navigating

In Paper Files, press `Space` to open Quick Look. While Quick Look is open, use ↑ and ↓ to follow adjacent files. Press `Space` again or close Quick Look to dismiss it.

Use the Quick Look window

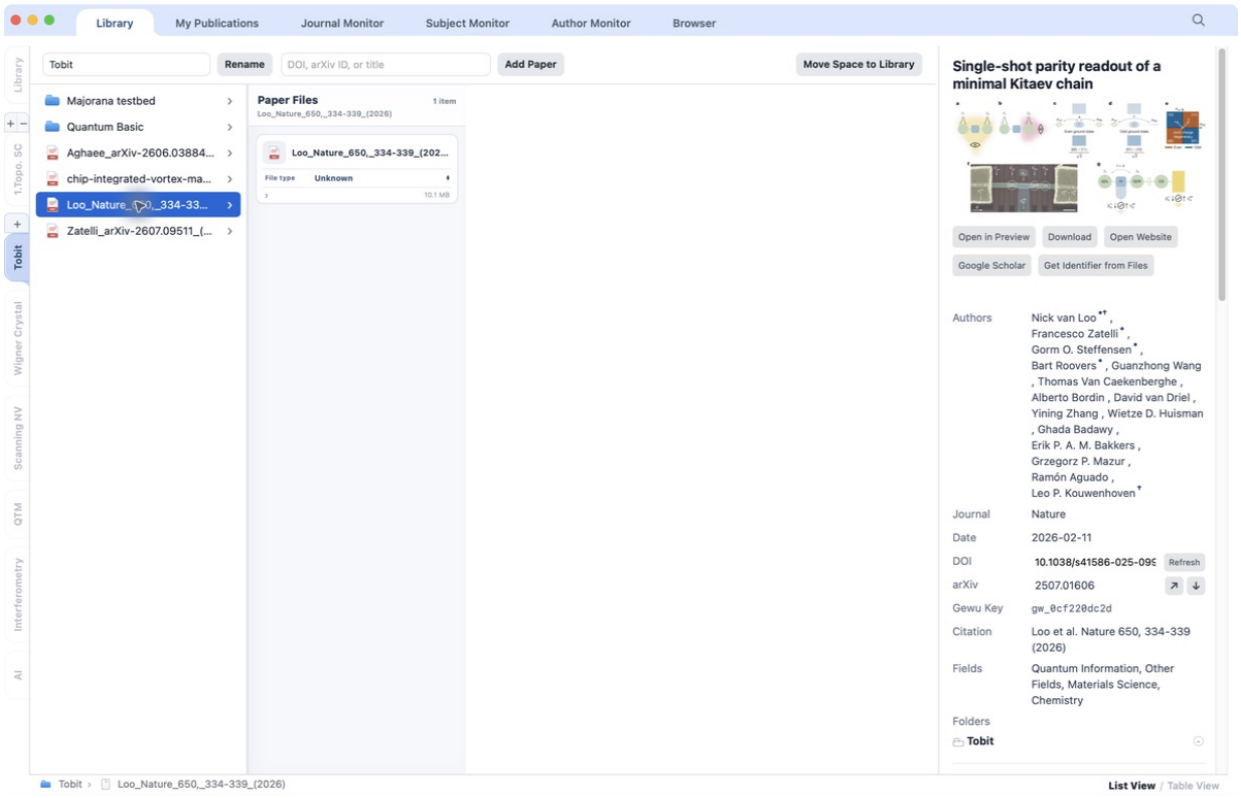
- For a PDF, use the thumbnail strip to jump to a page and scroll the page area independently.
- Drag the title bar to reposition the preview. Drag its resize corner when you need more reading space.
- Choose **Open in Preview** when you want the full macOS Preview application.
- When Quick Look is already open, changing the Paper Files selection updates the preview without requiring another open action.

> **Note**
> Single-click selects a Paper Files item. Double-click opens it. This prevents an accidental external app launch while you are classifying files.

See also: Understand Paper Files · Preview PDFs with Quick Look · Keyboard Shortcuts

3-1.3 Folders, Shortcuts, and Spaces

Folders organize a root. Shortcuts provide another entrance to the Library. Spaces keep an independent root for a project or research context.



A Space. Spaces use the same paper identity and Paper Files model as the Library while keeping an independent organizational root.

Choose the right structure

Structure	Use it for	What it changes
Library folder	A durable topic hierarchy inside the main collection.	Creates or moves a managed path.
Shortcut tab	Fast access to a Library location you use often.	Adds another entry point; it does not duplicate the paper record.
Space	An independent project, course, collaboration, or temporary collection.	Uses its own managed root under data / Spaces.

Create, rename, or move

1. **Control-click the relevant folder area.** Choose **New Folder**, **Rename**, **Copy**, or **Paste**.
2. **For a move, drag the item to its destination.** Wait for the destination to indicate it can receive the item, then release.
3. **Check the breadcrumb and Paper Details.** The visible path and folder membership update immediately.

Shortcut and Space tabs can be dragged to change their order. Their inactive colors distinguish the tab groups; the active tab uses the common selected style. With the keyboard, press ← from the leftmost content column to focus this rail, use ↑ and ↓ to open another source, then press → to enter its content.

Work with several entries

Command-click to add or remove individual entries from a selection. Shift-click, or hold Shift while using the arrow keys, to extend a range in the current folder. Copy the selected entries with `⌘C`, paste into the active destination with `⌘V`, or press `Delete` to apply the normal reference-aware deletion rules.

Copy a paper to another folder

A Library copy creates another reference to the same paper record rather than a second unrelated identity. The local filesystem may use a secondary symbolic link where appropriate. Paper versions and attachments remain associated with the work.

Move an entire Space into Library

Use this when a temporary or project Space should become part of the permanent Library hierarchy.

1. **Select the Space on the left rail.** Open the Space that you want to keep as a normal Library folder.
2. **Choose Move Space to Library.** The move sheet explains that the Space will become a folder with the same name.
3. **Choose Destination in My Library.** Select the Library folder that should contain the moved Space.
4. **Confirm the move.** The complete Space hierarchy, including its files and subfolders, moves together.

When the move finishes, the Space tab is removed and Library opens the moved folder. Canonical paper identities, folder placements, and Paper Files resource associations remain connected; the operation does not create duplicate paper records.

> **Important**
> Move managed papers in Gewu Insight, not directly in Finder. The app updates the visible file, associated bundle, and live database paths together.

See also: How Gewu Insight Stores Files · Delete and Recover Items

3-1.4 Search, List View, and Table View

Search locates items across the managed collection. List View emphasizes browsing; Table View emphasizes metadata comparison.

Search the collection

1. **Select the search control in the top-right corner.**
2. **Enter a title, author, journal, identifier, or filename.** Results may come from the Library and Spaces.
3. **Select a result.** Its Paper Details and managed path become available.
4. **Clear the search.** Return to the previous Library location.

Switch views

Choose **View > List View** or **View > Table View**. You can also use `⇧⌘V`. Table columns can be sorted or resized where available; changing the view does not rename, move, or rewrite files.

Sort a folder

Control-click the folder contents and choose **Sort by Name** or **Sort by Modified Time**. Sorting changes presentation, not the paper identity.

See also: Library Layout · Keyboard Shortcuts

3-2 Papers and Files

3-2.1 Add Papers and Files

A normal Library import creates a managed local copy and a database record. The original source file remains where it was.

Drag a paper into a folder

1. **Open the destination folder in Library.**
2. **Drag a PDF from Finder into the item area.** Drop it on the folder or in its contents.
3. **Wait for the item to appear.** The app copies the file into managed storage and begins local identification.
4. **Review Paper Details.** Confirm the title, DOI, arXiv identifier, and file classification.

Add a paper by DOI, arXiv ID, URL, or title

1. **Open the destination folder.** The resolved paper will be placed in the active Library folder.
2. **Choose Add Paper.** Control-click blank space in the folder contents and choose the command to open the inline lookup field.
3. **Enter a DOI, arXiv identifier, supported article URL, or title.** A DOI or arXiv identifier is the strongest identity clue. A title search may return alternatives that need review.
4. **Review and confirm the resolved match.** Check the title, authors, journal, and year, then choose **Add Paper** again.

If the identity already exists, Gewu Insight reuses the canonical paper and adds the selected folder placement rather than creating an unrelated duplicate.

> **Note**
> A paper record does not require a local PDF. **Add Paper** creates or reuses the metadata record; it does not automatically download a PDF. Use **Download** or **Add File** separately when you want a managed local file.

> **Note**
> A file added to an existing paper's **Paper Files** bundle is an attachment or version of that paper. A file dropped into a normal Library folder is initially its own Library item.

See also: Add and Classify Attachments · Use Folder Loader · Move a Standalone File into Paper Files

3-2.2 Use Paper Details

Paper Details combines actions, identifiers, metadata, and citation counts. Library, My Publications, and Author Monitor start with the paper title, without a separate heading bar. Action labels stay on one line; whole buttons wrap when space is narrow.

Top actions

Action	What it does
Open in Preview	Opens the preferred local PDF in macOS Preview.
Download	Retrieves the best available paper file without overwriting a different preserved version.
Open Website	Opens the publisher, DOI, arXiv, or best known work page in a Browser tab.
Google Scholar	Searches for the work in Google Scholar.
Get Identifier from Files	Examines local paper files for DOI or arXiv evidence.

File management

Library uses the adjacent **Paper Files** column; Paper Details does not repeat a Files box. My Publications keeps its collapsible **Files · n** management panel. Author Monitor shows a compact local filename with the full path on hover; **Get Identifier from Files** appears when a local PDF is available.

Metadata rows

The title appears once at the top. The rows below include authors, journal, date, DOI, arXiv, Gewu Key, citation, fields, and folder membership when available. **Gewu Key** is the stable, human-visible storage handle for a cataloged work record. It is not a DOI, arXiv identifier, or the database's relational primary key.

Compare citation sources

Merged Citations brings together the counts returned by OpenAlex, Semantic Scholar, Crossref, NASA ADS, and PubMed when those services have a match. Select a source label to inspect that source's count and evidence. The headline is a merged best estimate; counts from different services are not added together.

Inspect influential citations

When Semantic Scholar reports an **Influential** count, select it to inspect citing papers that Semantic Scholar classifies as especially meaningful to the cited work. This is Semantic Scholar's classification, not a score calculated by Gewu Insight. The list can include citation context, intent, citing authors, venue, year, and citation counts when the service supplies them.

Influential Citations is fetched on demand and cached locally for several hours. A saved API key gives Gewu Insight an individual Semantic Scholar allowance, but it is still limited to about one request per second and can be temporarily throttled. If the service returns a rate-limit message, the app pauses repeat requests for a few minutes; the paper record, existing citation totals, and any previously displayed result remain unaffected.

Read and reuse the record

- A representative first figure, when available, provides a visual checkpoint above the actions.
- **Abstract** keeps the paper summary in the independently scrolling details pane.
- **BibTeX** and **Copy Code** provide a citation record for reference software or LaTeX.
- **Nature Reference** and **Copy Reference** provide a formatted reference string.
- **Folders** lists every Library location for the shared work. Remove a location there only when you intend to remove that reference.

> **Note**
> Gewu Key is assigned when a work is first cataloged in SQLite, before Add to Library. Its row remains visible in Author Monitor, including while details load; a missing value shows `(undefined)`. Adding to Library reuses the existing key and updates the view without creating a second identity.

See also: Edit DOI and arXiv Identifiers · Understand Paper Files · How Gewu Insight Stores Files

3-2.3 Edit DOI and arXiv Identifiers

DOI and arXiv can identify the same work across publication stages. Correct either value in Paper Details before downloading or synchronizing versions.

Refresh or correct a DOI

1. **Select the paper.** Find the **DOI** row in Paper Details.
2. **Select the DOI field.** Enter or paste the correct DOI. You can omit a leading `https://doi.org/`.
3. **Choose Refresh or press `Return`.** Gewu Insight saves the identifier, checks supported metadata services, and updates matching bibliographic fields.
4. **Inspect the result.** Verify the title, journal, year, and authors before downloading a new file.

Edit an arXiv identifier

1. **Select the arXiv value.** The value changes to an inline text field.
2. **Enter an identifier.** Use forms such as `arXiv:2106.14849` or a valid legacy identifier.
3. **Press `Return`.** The database saves the new value. Enter an empty value and press `Return` to clear it.

The **arXiv** row remains visible when empty so you can add a corrected identifier later. Use the arrow icon to open arXiv and the download icon to retrieve the latest available arXiv PDF.

> **Important**
> Confirm an identifier before running a version sync. A wrong identifier can attach files from a different work.

See also: Download and Sync Versions · Open Source Settings

3-2.4 Understand Paper Files

A paper is one work record that can own several PDFs and related files. Paper Files shows that bundle without pretending it is another Library folder level.

Compact cards for one flat bundle

The Paper Files column is a flat peer list, not a hierarchy of version or attachment folders. Each card shows its filename, **File type**, and size by default. Click the size footer or its chevron to expand **Annotation** and **Source** into separate rows; click again to collapse them. Each card expands independently, and the choice lasts for the current session. File type remains visible because it identifies a preprint, accepted manuscript, published version, supplement, or other resource. The **Default** marker identifies the preferred file.

Select, open, or act on a file

- Single-click a card to select it.
- Double-click a card to open it.
- Double-click the paper row to the left to open the paper's Default file, independently of the highlighted resource card. PDFs open in macOS Preview. If an old unclassified import has no Default, its available current file still opens.
- Press `Space` to use Quick Look.
- Control-click for **Open**, **Quick Look**, **Show in Finder**, default-file controls, type controls, add/sync actions, or deletion where allowed.
- Press `Delete` or `Backspace` while Paper Files has keyboard focus to remove only the selected removable resource after confirmation. This does not delete the parent paper or its other files.

The physical primary file used by older compatibility paths remains protected from direct deletion. Choosing **Set as Default** makes that resource the file used by **Open in Preview** and paper preview without moving, renaming, or rewriting it. If you set it manually, **Use Automatic Selection** returns control to the file-type priority rule.

Library Paper Details no longer repeats this bundle. In My Publications, which has no adjacent Paper Files column, the collapsible **Files** panel keeps its file-management controls. **Show in Finder** reveals a physical file.

Deleting a private managed attachment moves its bytes to macOS Trash before removing its resource record. Shared files and external originals are preserved. A failed database deletion restores the moved file when macOS returns its Trash path. The interface shows progress and updates only after success; path indexes keep the shared-file safety check fast even in large libraries.

Understand a Missing File row

Missing File means that SQLite still has a resource record, but the file is not present at the recorded local path. Gewu Insight keeps genuine missing resources visible so you can identify and repair them; it hides only an older untyped migration alias when it can prove that the alias has already been replaced by the same paper's live canonical file.

A missing attachment is not another copy of the default PDF merely because both rows have a similar name. Check **Show in Finder**, the paper's Associated Files bundle, and Recycle before deleting or importing

anything. The default resource and explicitly typed attachments remain visible even when their paths need repair.

See also: Preview Files with Quick Look · Choose File Types and the Default File · Add and Classify Attachments · Navigate with Mouse and Keyboard

3-2.5 Preview Files with Quick Look

Press Space to inspect the selected resource without changing its paper record. Quick Look can stay inside Gewu Insight, use the macOS system panel, or combine both approaches.

Open and follow a preview

1. **Select a file in Library, a Space, or Paper Files.** Press `Space` to open the configured Quick Look.
2. **Move with ↑ or ↓.** When Quick Look is open, the preview follows the current Gewu Insight selection.
3. **Press Space again.** Close Quick Look and return keyboard focus to the same file.

With macOS Quick Look, Gewu Insight keeps keyboard focus so you can continue moving through Library or Paper Files while the system panel follows the selection. Rapid navigation is serialized so an older preview request cannot replace a newer selection.

Read the built-in PDF loading sequence

The document viewer starts first and requests the bytes it needs from the local file instead of waiting for the whole PDF to be copied into memory. Page count and other metadata load in parallel. Page thumbnails begin after the main document is ready, so a large PDF can already be readable while the thumbnail strip is still filling in.

On the first open, the app may still need to inspect the PDF and render thumbnail pages. Later opens can reuse stable local-file validators, remembered page information, and a bounded thumbnail cache. The cache is automatically trimmed and can always be rebuilt; it is not part of the paper record or a substitute for the PDF.

Choose Quick Look or Preview

Use	Best for
Built-in Quick Look	Fast PDF inspection, keyboard comparison among versions, and a preview that remains inside the workspace.
macOS Quick Look	System-native preview support for presentations, media, images, PDFs, and other formats supported by macOS.
Open in Preview	Full macOS Preview tools, annotation, export, printing, and a separate reading window.

> **Note**
> Previewing is read-only from Gewu Insight's perspective. Opening Quick Look, loading thumbnails, or moving between pages does not change file type, annotation state, source, Default status, or the SQLite paper record.

See also: Quick Look Settings · Navigate with Mouse and Keyboard · Understand Paper Files · Troubleshooting

3-2.6 Choose File Types and the Default File

The default file is the preferred local representation of the paper. You can choose it manually or let Gewu Insight select it automatically.

Automatic selection

The automatic order is **Corrected published version**, **Published version**, **Accepted manuscript**, then **Preprint**. Within a type, a later recognized version normally wins. Correction notices, supplementary material, peer review files, Other, Unknown, and Invalid file never become the automatic default.

Automatic and manual eligibility

Only the four publication types above participate in automatic selection. You may manually choose any existing file except one classified as **Invalid file**. A manual choice is sticky across rescans, metadata refreshes, and later downloads until you choose **Use Automatic Selection**. Newly arriving files remain peer rows and do not erase or silently reclassify older files.

Choose a file manually

1. **Enter Paper Files, or expand Files in My Publications.**
2. **Select the file you want.**
3. **Choose Set as Default.** This file will be used by **Open in Preview** and paper preview.

> **Result.** The selected file becomes the manual default. Its existing filename and physical location are preserved.

Return to automatic selection

Choose **Use Automatic Selection** for a manually selected default. Gewu Insight evaluates the eligible file types again.

Classify files without collapsing annotation state

File type stays visible on each card. Click the size footer to reveal Annotation and Source, then change each control independently. File type uses exactly these labels:

- **Preprint**, **Accepted manuscript**, **Published version**, and **Corrected published version**
- **Correction notice**, **Supplementary material**, and **Peer review file**
- **Other**, **Unknown**, and **Invalid file**

Annotation state is selected independently as **Annotated**, **Unannotated**, or **Unknown**. Source is independently recorded as **Publisher**, **Preprint server**, **Institutional repository**, **User import**, **Other**, or **Unknown**. Manual type and annotation choices survive later fingerprint rescans.

Publisher is used only when a recognized publisher host or a publisher-specific acquisition path provides that evidence. arXiv, bioRxiv, and medRxiv are shown as **Preprint server**; repository hosts remain **Institutional repository**; ordinary local additions are **User import**; and a generic web download with no stronger evidence remains **Other**. A real host can appear after the source label when useful. Internal migration or ingestion markers are never user-facing source details, and choosing a different source clears a stale provider detail rather than appending it to the new choice.

> **Note**
> Changing a type, annotation state, source, or default changes database metadata only. It does not rename the file merely to match the label.

See also: Download and Sync Versions · Add and Classify Attachments

3-2.7 Add and Classify Attachments

Import first, classify afterward. This keeps the add operation simple and lets you set type, annotation state, and source after seeing the file in context.

Add a file to the paper

1. **Control-click the Paper Files area and choose Add File.** In My Publications, expand Files and use Add File.
2. **Select one or more local files.** Confirm the file chooser.
3. **Wait for the new row.** A new item begins as **Unknown** unless its source proves a more specific type.
4. **Select its compact controls.** Choose File type; click the size footer to expand Annotation and Source when needed.

Attachment types

The same ten file types apply to every row. File extension and MIME type remain separate technical facts, so an image, note, dataset, or code archive can use **Other** without being mistaken for a PDF version.

Add by drag and drop

Drag a Finder file over the expanded Paper Files area, wait for the drop target to respond, then release. The file is copied into the selected paper's associated bundle and appears in Paper Files.

> **Important**
> Invalid or unreadable PDFs are rejected at file ingress. Existing migrated records that can be proven invalid remain visible as **Invalid file** for audit and repair, but cannot become a default.

See also: Move a Standalone File into Paper Files · How Gewu Insight Stores Files

3-2.8 Download and Sync Versions

Downloads add a version to the paper bundle. They do not silently overwrite a different preserved PDF.

Download the best available paper

1. **Verify DOI and arXiv.** Make sure the identifiers belong to the selected work.
2. **Choose Download at the top of Paper Details.** Gewu Insight first tries the best available published file, then supported alternatives.
3. **Wait for the status message.** Existing files are preserved if the incoming file is a different version; a manual default stays selected.
4. **Inspect Paper Files.** Every preserved file appears in the same flat list. Automatic ranking changes only the Default marker; a manual default remains unchanged.

Download arXiv files

- Use the download icon in the **arXiv** row to retrieve the latest arXiv PDF.
- Choose **Sync Versions** to retrieve all available arXiv revisions such as v1, v2, and v3.
- Each revision is labeled with its arXiv version when known.

Duplicate protection

Gewu Insight computes exact SHA-256 and PDF content fingerprints as each new resource is registered. An exact duplicate is reused instead of stored twice. A clean PDF and a genuinely annotated copy can share a content fingerprint while retaining different exact hashes, annotation counts, and labels.

Validation before replacement

A response is not accepted merely because it begins with `%PDF-`. Structural PDF validation rejects empty files, HTML responses, truncated downloads, missing valid endings, and content that cannot be parsed as a PDF. A download is written to a temporary file and replaces the destination only after validation succeeds. If the request or stream fails, an existing valid file remains untouched and a partial file is not registered as a resource.

Classification follows the file that actually arrived

A publisher download is classified from the file and source actually returned. A corrected published version outranks a published version automatically; a publisher reference, First Release, or accepted preview remains an accepted manuscript. A correction notice remains a separate peer and never becomes automatic merely because it came from a publisher.

A PDF URL stored in bibliographic metadata is only a retrieval lead. It does not create a Paper Files row until a download actually succeeds, the returned bytes pass validation, and a local resource is registered.

If a download is skipped because the local bytes already exist, that skipped operation does not relabel the existing file from metadata or a newly discovered URL. Correct the file type explicitly if the saved classification itself is wrong.

> **Note**
> A Monitor result is not forced into a Library placement. Where download is available, a local resource can belong to the cataloged paper even before you add that paper to Library or My Publications.

See also: Choose File Types and the Default File · Publisher and Network Access · Edit DOI and arXiv Identifiers

3-2.9 Move a Standalone File into Paper Files

If a data, image, presentation, or other file was imported as its own Library item, drag it onto the correct paper's Paper Files area to make it an attachment.

Convert an independent item

1. **Select the destination paper.** Keep its Paper Files inspector visible.
2. **Begin dragging the standalone item.** The main paper remains the active destination while you drag.
3. **Move the pointer into the Paper Files drop zone.** Wait for the drop feedback that identifies the target paper.
4. **Release the pointer.** Gewu Insight validates the move, adopts the file into the paper bundle, and removes its independent Library position.
5. **Classify the attachment.** Choose **Supplementary material** for an actual paper supplement, or **Other** for a general image, dataset, code file, slide deck, or note. Expand the size footer to set Annotation and Source separately. See the complete current file-type list.

> **Result.** The physical file and database link now belong to the destination paper; the former standalone entry is gone.

If the conversion cannot finish

The file moves and database changes are coordinated as one recoverable operation. If any step fails, Gewu Insight restores the original standalone files and rolls back the paper, Library item, resource relationships, and fingerprint fields. The source should remain usable rather than becoming a half-converted attachment.

> **Important**
> If the source item is shared by another Library reference or protected relationship, Gewu Insight may refuse a destructive conversion. Add a copy with **Add File** instead.

See also: Add and Classify Attachments · Folders, Shortcuts, and Spaces

3-2.10 Delete and Recover Items

Deleting a paper reference either removes only that reference or moves the last Library reference to Recycle. Final deletion inside Recycle is reference-aware and uses the macOS Trash for managed files.

Delete a Library reference

1. **Control-click the item.**
2. **Choose Delete.**
3. **Check the result.** If another Library location or My Publications still refers to the same work, only the selected reference is removed. Otherwise the item moves to **Recycle** and remains recoverable.

Recover an item

1. **Open Recycle.**
2. **Control-click the item and choose Recover.**
3. **Confirm the restored location.**

Delete permanently

Use final deletion only inside Recycle. The confirmation summarizes how many paper records will be removed, identifies managed local copies that will move to the macOS Trash, and states that original imported source files outside Gewu Insight are not touched. If the same paper is still referenced by another folder or My Publications, the shared record and files are preserved.

> **Important**
> **Delete All** affects every eligible item in the selected Recycle context. Back up first if you are uncertain.

See also: Back Up or Move to Another Mac · How Gewu Insight Stores Files

3-3 Import and Transfer

3-3.1 Use Folder Loader

Folder Loader imports an existing directory tree into managed Library storage while preserving relative paths.

Import a folder

1. **Choose Loader > Open Loader.** Outside Browser, you can also press ⌘L; inside Browser that shortcut focuses its address field.
2. **Choose the source folder.** Review the displayed path before starting.
3. **Select a metadata mode.**
   - **Skip metadata** — fastest; import files and structure only.
   - **Quick refresh** — recommended for most migrations.
   - **Full refresh** — more network work and a longer run.
4. **Choose Import.** Follow progress in the live log. Use **Stop Import** if necessary.
5. **Review the result.** Open previous logs or the result guide when items need attention.

What Folder Loader preserves

- The source folder's relative hierarchy.
- Existing managed targets and files on repeated imports.
- An existing primary PDF rather than overwriting it.
- A verified exported Paper Files relationship when a Gewu Insight manifest is present.

Folder Loader copies into Gewu Insight's managed storage. It does not move, rename, or delete the external source files and folders you selected.

Delete an import batch

Delete Import is a rollback for the selected Loader batch, not a way to forget a log. Stop an active import first. The command removes database records created by that batch, eligible empty folders, its Loader reports, and managed local copies associated with the batch; eligible paths are sent to the macOS Trash. It never deletes the original external source directory.

> **Important**
> Review the selected batch and make a backup before using Delete Import, especially if you have already reorganized or reused imported items.

> **Note**
> For an ordinary folder, Loader does not guess which files are "primary" versus "associated." It imports the collection as supplied. Special reconstruction occurs only for a verified Gewu Insight export manifest.

See also: Make a Portable Desktop Copy · Refresh an Imported Library · Back Up or Move to Another Mac

3-3.2 Refresh an Imported Library

Loader maintenance commands fill metadata in batches after the files and folder structure are safely imported.

Refresh known works

1. **Choose Loader > Refresh All Metadata.**
2. **Let the queue run.** Eligible Library papers with DOI or arXiv identifiers are refreshed sequentially.
3. **Review the log.** Recently successful records may be skipped to avoid unnecessary requests.

Find missing identifiers

1. **Choose Loader > Find All Missing Identifiers.**
2. **Wait for supported services.** The app checks active Library and Space PDFs using Crossref, OpenAlex, and Semantic Scholar where available.
3. **Review uncertain matches.** Refresh or recover DOI evidence as needed, and edit an incorrect arXiv identifier inline before downloading versions.

Verified "not found" checks and transient failures use cooldowns so a repeated run does not immediately send the same requests. Bulk missing-identifier work does not use Google Scholar.

See also: Open Source Settings · Publisher and Network Access · Edit DOI and arXiv Identifiers

3-3.3 Make a Portable Desktop Copy

Make Copy Desktop creates a self-contained copy for transfer, inspection, or re-import.

Create a copy

1. **Control-click the non-root Library folder you want to export.**
2. **Choose Make Copy Desktop.**
3. **Wait for the completion message.** Open the resulting Desktop folder in Finder.
4. **Confirm representative files.** Check a paper with versions or attachments as well as a normal folder.

What is included

The export replaces managed filesystem shortcuts with real files, includes relevant Associated Files, and writes a verified manifest describing paper-file relationships. This lets Folder Loader restore the bundle on another installation without guessing from filenames.

> **Important**
> Do not edit or discard the export manifest before migration. Without it, Loader correctly treats the files as an ordinary folder and does not infer hidden associations.

See also: Use Folder Loader · Back Up or Move to Another Mac · How Gewu Insight Stores Files

3-3.4 Back Up or Move to Another Mac

A reliable transfer preserves both the SQLite database and the managed file roots. The app bundle alone is not your library.

Create a full backup

1. **Quit Gewu Insight with ⌘Q.** Wait until background refresh or download work has stopped. Closing a window is not the same as quitting; also stop any separately launched local UI or CLI process before copying its database.
2. **Copy the complete project directory.** This is the safest recovery unit because it keeps `data`, `reports`, `config`, the app source, and managed paths together.
3. **Preserve directory names and permissions.** Do not separate SQLite from the managed file roots it describes.
4. **Store the backup separately.** Use a second disk or a versioned backup location.

Back Up and Upgrade

1. **Quit the current version and make a full backup.** Preserve SQLite and every managed file root together.
2. **Install or build the newer application.** A normal software upgrade does not overwrite the existing `data` directory.
3. **Start the new version once.** Required database schema changes run automatically against the existing SQLite database.
4. **Verify before deleting the backup.** Open the default PDF, another Paper Files version, an attachment, and a My Publications entry.

> **Note**
> Schema migration updates database structure; it is not a reason to rebuild the database or re-import the Library. Keep the pre-upgrade backup until the new version has been checked.

Move to another Mac

1. **Install or update Gewu Insight on the destination Mac.**
2. **Keep the destination app closed.**
3. **Restore the matching database and managed file roots together.** Do not mix a new database with an unrelated old Associated Files tree.
4. **Start the app and test representative works.** Open a paper, an older version, and an attachment; verify My Publications and one monitor.

Transfer only part of a Library

Use **Make Copy Desktop** on the source Mac, move the complete exported folder, then use Folder Loader on the destination Mac. Keep its verified manifest intact.

Browser and credential backup boundaries

The project database includes Browser bookmarks, folders, history, and saved sidebar preferences. A bookmark HTML export transfers only bookmarks, not history or research papers. Open webpage tabs and unsubmitted forms are memory-only and are not restored after quitting.

WebKit website cookies, sign-ins, and cache belong to a macOS-managed Browser profile outside this project. Copying the project or a bookmark export is not a supported transfer of that profile; expect to sign in again on another Mac. Apple Passwords and its passkeys remain under macOS/iCloud management and are never part of a Gewu export. Optional metadata API keys saved in SQLite do travel with the database: keep full backups private.

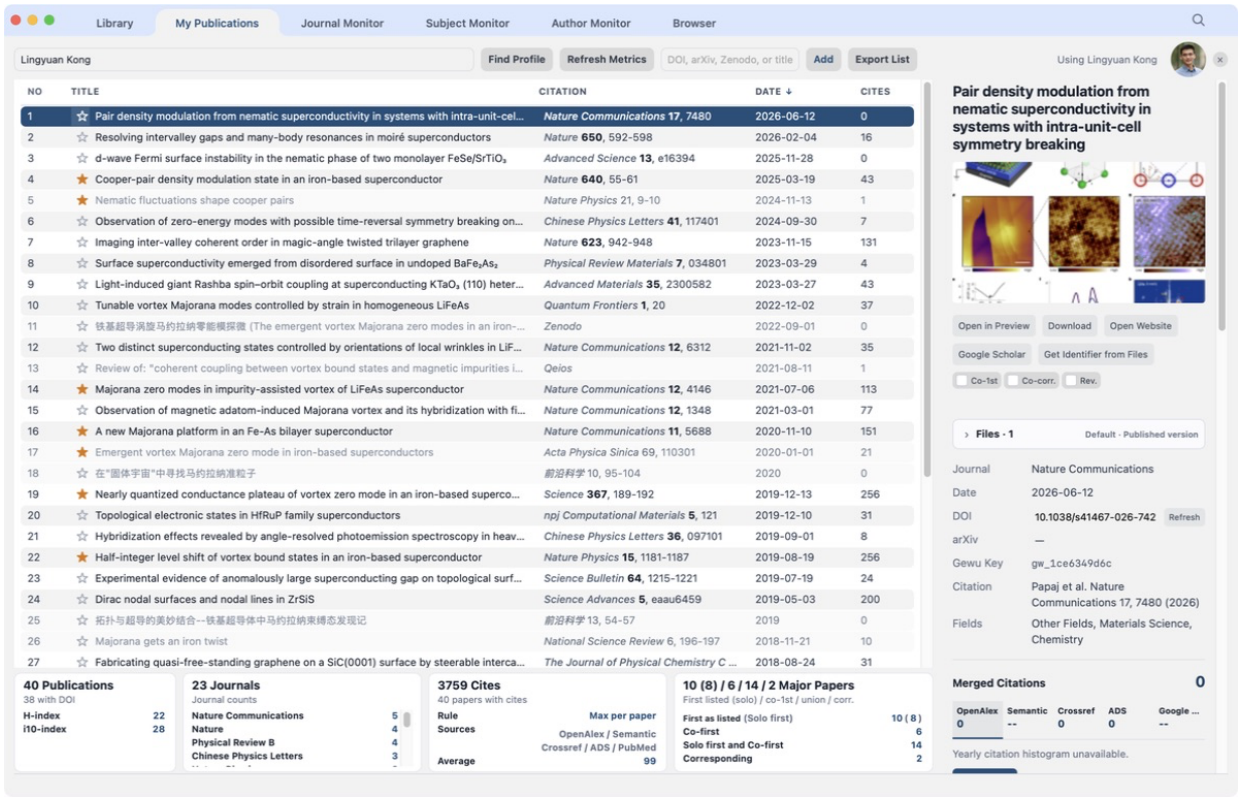
> **Important**
> Never migrate by copying only `paper_monitor.sqlite` or only the visible Library folder. Either choice leaves half of the managed state behind.

See also: Make a Portable Desktop Copy · Storage Reference · Troubleshooting

3-4 My Publications

3-4.1 My Publications

My Publications keeps a managed publication list, citation sources, author-role statistics, and export tools for your own work.



My Publications. The active profile, DOI-resolved publication list, citation evidence, article figure, and authorship summary share one view.

Choose your profile

1. **Enter a name, optional affiliations, or an exact Google Scholar profile URL.** For a name search, use `Name; affiliation A; affiliation B`.
2. **Choose Find Profile.** Review candidate profiles carefully.
3. **Choose Use Profile.** The saved strip shows the active name and profile photo. Select the photo to open the profile in Browser; use the remove control to forget it.

Add and update publications

- Use the identifier/title field and **Add** to add a DOI, arXiv record, supported identifier, or title.
- Choose **Refresh Metrics** to update citation services and PubMed data.
- **First as listed** is calculated automatically from the active profile name and the resolved author order, including the profile's English and Chinese name forms.
- Use **Co-1st**, **Co-corr.**, and **Rev.** in Paper Details to mark co-first authorship, corresponding authorship, and review/perspective papers. Leave Co-1st unchecked for a solo first-author paper. These manual roles are preserved during metric refresh.
- Use the star to highlight a paper. Star state is independent from citation refresh and export inclusion.
- Drag rows into a manual order when a curated sequence matters more than date or citation sorting.
- Choose **Comments** to build or open a citation-comments report from citing-paper metadata. This is not a personal note field.

- Use **Hide** or **Show** in the context menu to control export inclusion. Use **Open in Finder** or **Delete** for the corresponding local action.

Build a citation-comments report

Comments gathers citing-paper metadata and available Semantic Scholar citation-context sentences, then writes a local report that groups strong praise, positive impact statements, cautions, and unmatched citing records. It helps you review how later literature describes a publication; it does not modify the publication metadata or replace your own notes.

The report is a generated local reading artifact. Network services may return fewer contexts than citation records, and a temporary rate limit can leave some entries without quoted context. Reopen or rebuild the report later to retry the available sources; an existing report remains on disk until you deliberately replace or remove it.

Work with files

My Publications uses the same Paper Details and flat Paper Files model as Library. Expand **Files · n** within the details to inspect its resources; Library instead has an adjacent Paper Files column. A publication can keep a default PDF, earlier versions, supplements, notes, presentations, data, and other attachments under the same Gewu Key.

Read the summary

The lower summary includes publication counts, identifier coverage, h-index and i10 information, journal distribution, merged citation maxima, and major-paper role counts. Select a role statistic to filter the table where supported.

Export a list

Choose **Export List**, then select LaTeX PDF, LaTeX source, Overleaf project ZIP, or Word document. Papers excluded from export remain visible in My Publications but are omitted from the generated list. Exports use the saved manual order where that format supports it and never replace the SQLite source.

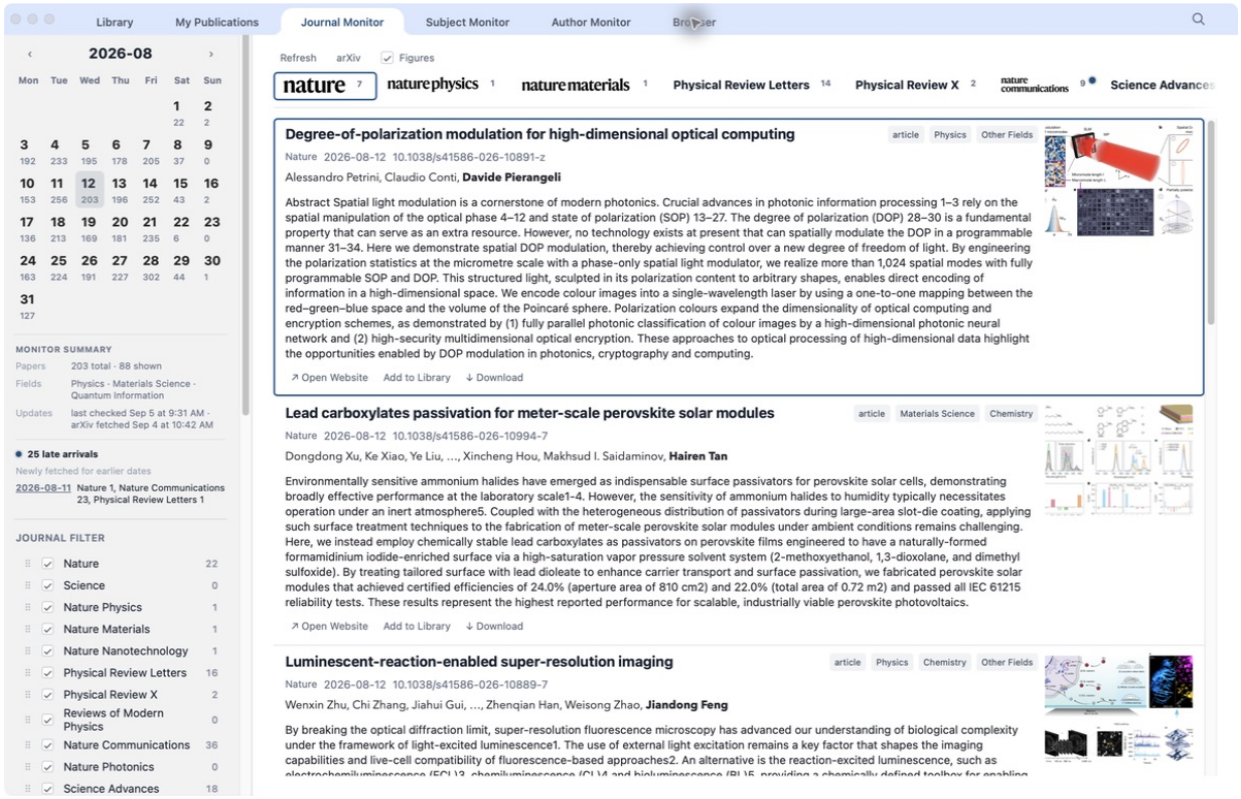
The Overleaf option downloads a project ZIP and opens Overleaf in Browser. It does not upload your files or create an Overleaf project automatically; import the ZIP yourself on the website.

See also: Open Source Settings · Use Paper Details · Local Data and Privacy

3-5 Journal Monitor

3-5.1 Journal Monitor

Journal Monitor collects papers for a selected publication date and lets you narrow the result by journal, preprint source, and field.



Journal Monitor. Nature is selected for August 12, 2026, with scientific figures visible beside the paper cards. The independent sidebar retains calendar counts, the monitor summary, and journal and field filters.

Refresh a date

1. **Select a day in the calendar.** Small numbers below a day show stored result counts.
2. **Choose Refresh.** The app requests all configured journal sources for that date.
3. **Use Refresh This Journal when only the active journal needs another attempt.** This avoids re-requesting journals that are already complete.
4. **Choose arXiv separately if needed.** This keeps journal and preprint work explicit.
5. **Watch the bottom status area.** It reports the current source and completion without changing the page height.

Manage monitored journals

Open **Journal Settings** to add or manage a journal by name, ISSN, or known publication history. When an ISSN is missing, the app can look up a matching journal record before saving it. Review the match rather than relying on a similar title alone.

Drag journals vertically in **Journal Filter** or **Journal Settings** to set their order. The saved order controls the filter list and the journal groups in the report. Removing a journal disables the subscription without erasing its previously collected papers or its saved history.

Your enabled list, disabled history, ISSNs, source identifier, section policy, logo source, and display order are saved in SQLite. Product updates can add better built-in source definitions, but those defaults do not replace this personal state. Adding a previously removed journal restores it at the end of the active order.

Understand counts and filters

The number beside each journal is the total stored count for that journal on the selected day. It is deliberately independent of **Field Filter**. The field checkboxes narrow the paper cards you can see; they do not rewrite the journal totals. A journal can therefore show a nonzero count while the current field selection hides every one of its papers.

Calendar counts summarize stored day results. A zero count is meaningful only after the relevant source completed successfully. Read the source status before interpreting zero as "the journal published nothing."

Daily scheduled updates

Journal Monitor is designed to keep a daily cached record through its scheduled update workflow. A scheduled run and a manual refresh use the same stored day results, so opening a date does not require downloading it again. Use a manual refresh when a scheduled source was missed, partial, or needs verification.

The scheduler belongs to the running local UI service; it does not wake a sleeping Mac or run while that service is stopped. See When automatic updates run for the current timing and restart boundaries.

Journal logos and mathematical text

Configured journal logos are downloaded into a local cache and reused from disk. The first successful request may require network access; changing dates and later launches should not repeatedly download the same logo. The cache is presentation data and can be rebuilt without changing subscriptions or papers.

Supported inline LaTeX in titles and abstracts is typeset after the math renderer becomes ready. On a cold launch, raw source may be visible briefly, then the current day is rendered again automatically; changing to another date and back should not be necessary. Title normalization also preserves recognized proper names and source capitalization such as *Su–Schrieffer–Heeger* and *Monte Carlo* while still cleaning all-uppercase metadata and formula spacing.

How discovery provenance is preserved

Each stored Journal result is related to the canonical paper, the fetch run, and the source that reported it. The run preserves the journal, requested date window, status, and timing. Seeing the same paper from Crossref and OpenAlex records both source observations against the same paper identity; seeing it again within one run does not create a duplicate relationship. Later runs remain separately traceable.

The same canonical paper can also acquire Subject Monitor and Author Monitor relationships. These discovery paths explain why the work is visible; they do not automatically add a Library placement or create a local Paper Files resource.

Filter and inspect

Use **Journal Filter**, **Preprint Filter**, and **Field Filter**. Journal and field choices combine to determine the visible cards, while the journal counts continue to describe the full selected day. Enable **Figures** to show a representative image when available. Selecting a result opens its report and actions.

Act on a result

- **Open Website** opens the publisher or best known record in a Browser tab.
- **Add to Library** begins the placement workflow described below; after that, **Show in Library** returns to the managed paper.
- **Download** retrieves the paper file. **PDF** opens an existing local PDF in Preview, and **Delete PDF** removes that managed download and resets its status.
- When a reading note exists, **Note** opens it in the right-side drawer. The list compresses but keeps its own scroll position. **Delete Note** removes the note and resets analysis status.

Add a Journal result to Library

1. **Choose Add to Library on the result.** The Library destination chooser opens without leaving the selected date.
2. **Choose the target Library folder.** This is the visible folder placement for the managed paper.
3. **Complete the add.** The card changes to **Show in Library**; use it to reveal the existing placement later.

The operation links the Journal discovery to the canonical paper, preserves its source and run provenance, and adds the chosen Library placement. It does not automatically download a PDF. Choose **Download** separately when you also want a local file. If the paper is already managed, **Show in Library** reveals it instead of creating a duplicate identity.

Per-paper status text and the bottom progress area make completed downloads and background work visible, which helps avoid repeating the same action.

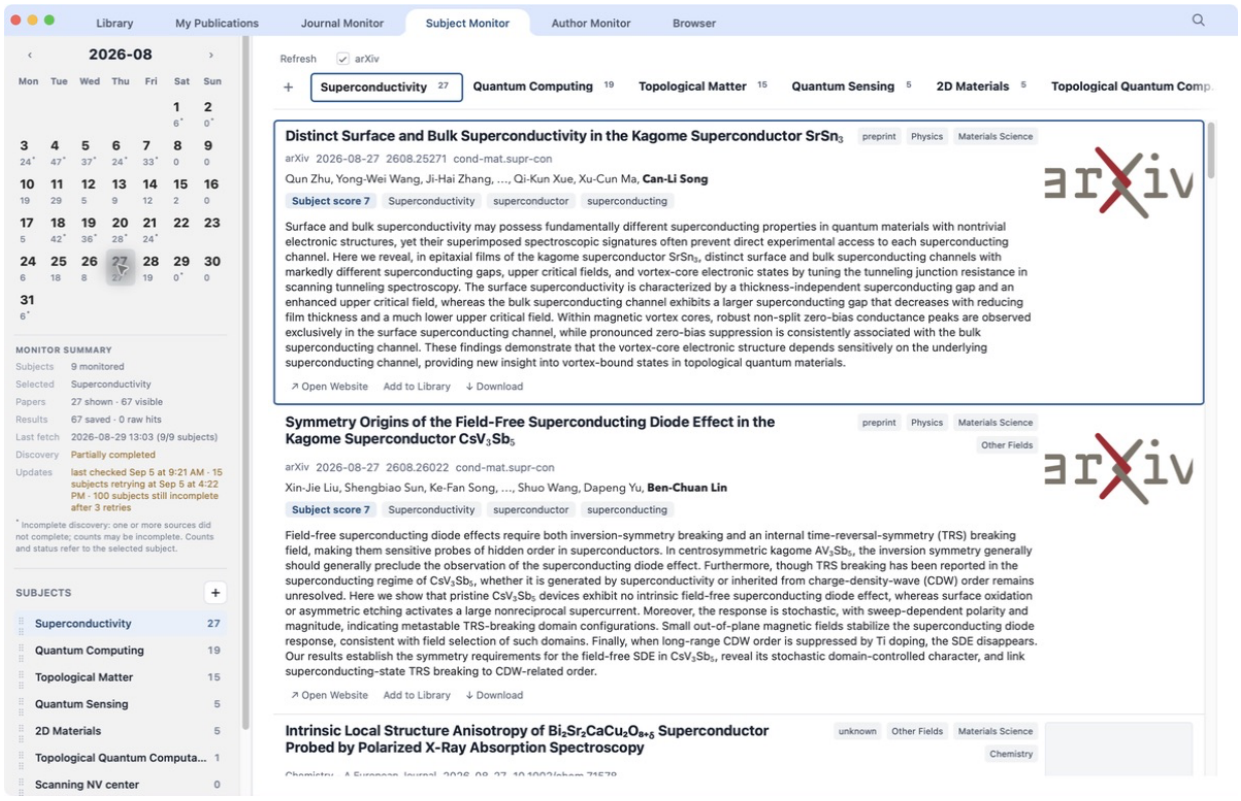
> **Note**
> "No matching papers" and "source fetch failed" are different outcomes. Check the source status before concluding that a date is empty.

See also: Subject Monitor · Publisher and Network Access · Add Papers and Files

3-6 Subject Monitor

3-6.1 Subject Monitor

Subject Monitor discovers papers for reusable topics. Each subject stores a name and literal discovery terms; an optional copied prompt can help you draft those terms outside the app.



Subject Monitor. Superconductivity is selected, with matching papers and a superscript asterisk on the partially completed count. The explanation sits below Updates at the bottom of Monitor Summary. Completed empty dates show an unmarked zero.

Create a subject

1. **Choose Add Subject.**
2. **Enter a clear name and discovery keywords.** Describe the field broadly enough to find variants, but avoid unrelated general terms.
3. **Use Copy Suggested Prompt if you want Codex-assisted wording.** Review the prompt before using it.
4. **Save and arrange the subject.** Subjects can be edited, deleted, and reordered.

Keywords use case-insensitive literal matching. Enter the words or phrases you expect to appear; they are not regular expressions and are not silently expanded into unrelated concepts.

Keep one subject selected across dates

Selecting a subject makes it the active topic until you choose another one. Moving to a date with no result does not switch the tab back to the first subject, and a subject with zero papers remains selectable. The calendar's day numbers describe the active subject, so you can see which dates contain papers for that topic rather than a total across every subject. The count beside each subject tab describes that subject on the selected date.

After all sources complete successfully for that subject and date, the calendar shows the stored count, including **0** for no matching papers. A partially completed fetch also shows its current count, including zero, with a superscript asterisk. The asterisk means one or more sources did not complete, so the count may be

incomplete; zero with an asterisk is not a confirmed empty result. A date with no stored results and no fetch record remains blank. While a fetch is running, the progress indicator remains visible.

The **Discovery** row describes the selected subject; its asterisk legend appears at the bottom of **Monitor Summary**, after Updates and any Errors. Discovery distinguishes **Completed**, **Partially completed**, failed, and deferred discovery; failed or deferred attempts also keep their provisional counts marked. Switching subjects changes both the counts and their completion markers. A later successful refresh removes the asterisk, and reopening the app preserves the status recorded in SQLite.

Refresh subjects

1. **Select a calendar date.** Selecting a date only loads the result already stored for that day; it does not start a network request.
2. **Choose Refresh.** The command refreshes every configured subject for the currently selected date. The arXiv display option is enabled by default; clear it when you want to hide arXiv cards without changing what is stored.
3. **Read source status.** A source can be complete, partial, empty, or failed. Source-level retries are managed by the background scheduler.
4. **Cancel if necessary.** Completed results remain distinguishable from unfinished work.

Refresh one subject

Control-click a subject and choose **Refresh This Subject** when only that topic needs another pass. Use the main **Refresh** command when every configured subject should run for the selected date.

The current candidate layer can draw from local arXiv data, local APS RSS data, OpenAlex, Crossref, Semantic Scholar, and PubMed. It does not depend on a fixed number of feeds. Results still pass topic filtering and deduplication before display.

Inspect and keep a result

Select a paper to read its title, authors, abstract, source, date, identifier, topic score, and matched subject terms. Use **Open Website**, **Download**, **PDF**, and the note controls in the same way as Journal Monitor. Choose **Add to Library** to create a managed reference, or **Show in Library** when one already exists.

Add a Subject result to Library

1. **Choose Add to Library on the matching paper.** The subject and selected date remain in place while the destination chooser opens.
2. **Select the target Library folder.** The same canonical paper can be placed in a durable topic hierarchy without losing the Subject relationship.
3. **Complete the add.** Use **Show in Library** on later visits to reveal the managed placement.

Adding the result preserves the matched subject, score, terms, and discovery provenance. The paper remains visible in Subject Monitor immediately after adding it; its action becomes **Show in Library**, while the selected subject, date, and reading position stay in place. It does not automatically download a PDF; use **Download** separately. Repeated discovery of the same identity updates relationships rather than creating unrelated paper records.

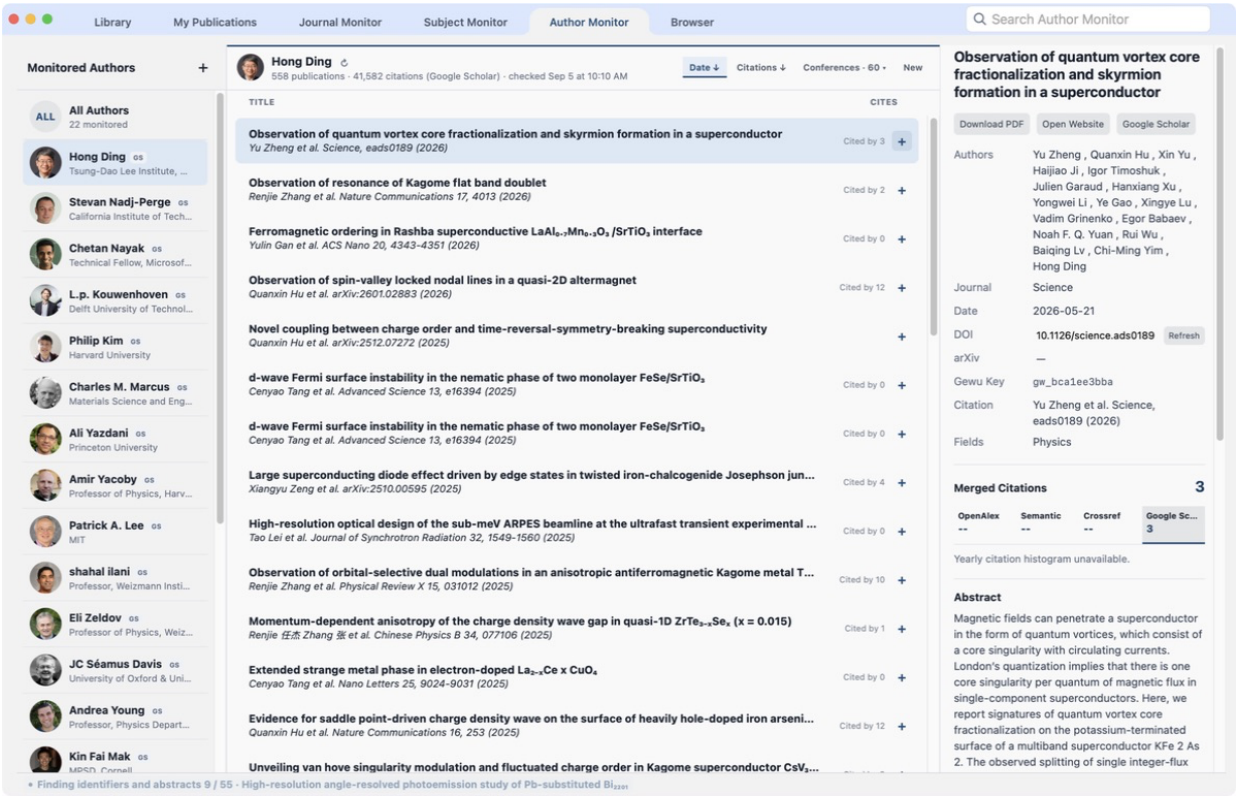
> **Important**
> Subject Monitor rejects Zenodo records during ingestion and when cached results are displayed. There is no Zenodo show/hide option.

See also: Journal Monitor · Publisher and Network Access

3-7 Author Monitor

3-7.1 Author Monitor

Author Monitor follows researchers and separates a first-import baseline from newly discovered work.



Author Monitor. Hong Ding's profile is selected. The paper title leads the details pane, followed by compact actions and a visible Gewu Key; local portraits keep the author list stable.

Add an author

1. **Choose the add-author control.**
2. **Enter a name and optional affiliations, or an exact profile URL.** Google Scholar and ResearchGate profile URLs are supported where the profile can be resolved.
3. **Confirm the correct person.** The initial result set becomes the baseline rather than a batch of unread "new" papers.

Understand profile and publication sources

A Google Scholar profile is refreshed through Google Scholar and follows the shared Scholar pacing and cooldown. A ResearchGate URL identifies the public profile, but its publication list is matched separately to a publication-data author. Gewu Insight uses OpenAlex for that match when it is configured and automatically falls back to Semantic Scholar when OpenAlex is unavailable. Confirm the proposed person rather than assuming similar names are identical.

If no publication-data match can be verified, the ResearchGate profile and homepage can still be saved, but automatic publication refresh remains unavailable until a match is supplied. The profile source, publication metadata provider, and per-paper citation source remain distinct evidence; a fallback does not silently turn the ResearchGate profile into a different author identity.

Review updates

Later refreshes can mark newly discovered publications as unread. Use the **New** filter and other author/result controls to focus the list. Conference or related popovers provide additional context where available.

Control-click **New** and choose **Read All** to clear the blue unread dots. With one author selected, this marks all new publications for that author as read, including those not currently scrolled into view; other authors keep their unread state, even for a shared paper. With **All Authors** selected, it marks all authors' new publications as read. The action preserves papers and Library placements. The New filter stays active, so newly read papers leave that filtered view; click New again to see all publications. Read state persists after reopening the app.

Update citation profiles

Update All Citations is an explicit full-profile operation. It walks the monitored profile and refreshes citation information paper by paper, so it can take a long time and should not be treated as a quick page reload. Start it only when you intend to update the whole profile, then let the bottom progress status finish.

Manage the profile image

Control-click an author and choose **Edit Homepage** or **Change Photo**. A downloaded Google Scholar avatar is saved locally, and local photo bytes are prepared when the service starts so opening Author Monitor does not need to reopen SQLite or contact the image host for every row. A manual photo takes precedence. If no valid local photo exists, the list can use initials until you explicitly refresh or replace the profile image.

Discovery entries and managed papers

An Author Monitor result is a cataloged shared record, with a Gewu Key shown in its details even before it has Library placements or local files. A missing key displays `(undefined)` until record data is available. Add to Library preserves the existing key and updates the view; downloading or attaching a file creates a resource relationship.

Add an author result to Library

1. **Select the publication and choose Add to Library.** This opens the Library destination chooser without removing the author from view.
2. **Choose the target Library folder.** The selected placement becomes the durable organizational path for the shared paper record.
3. **Complete the add.** The action becomes **Show in Library**, which reveals the managed paper later.

The author-publication relationship and canonical identity remain intact. Adding a placement does not automatically download a PDF, and a paper already managed elsewhere is reused rather than duplicated.

Remove a monitored author

Deleting an author from Author Monitor removes the monitoring relationship and its monitor state. It does not delete papers already added to Library, their managed PDFs, or their Paper Files bundles.

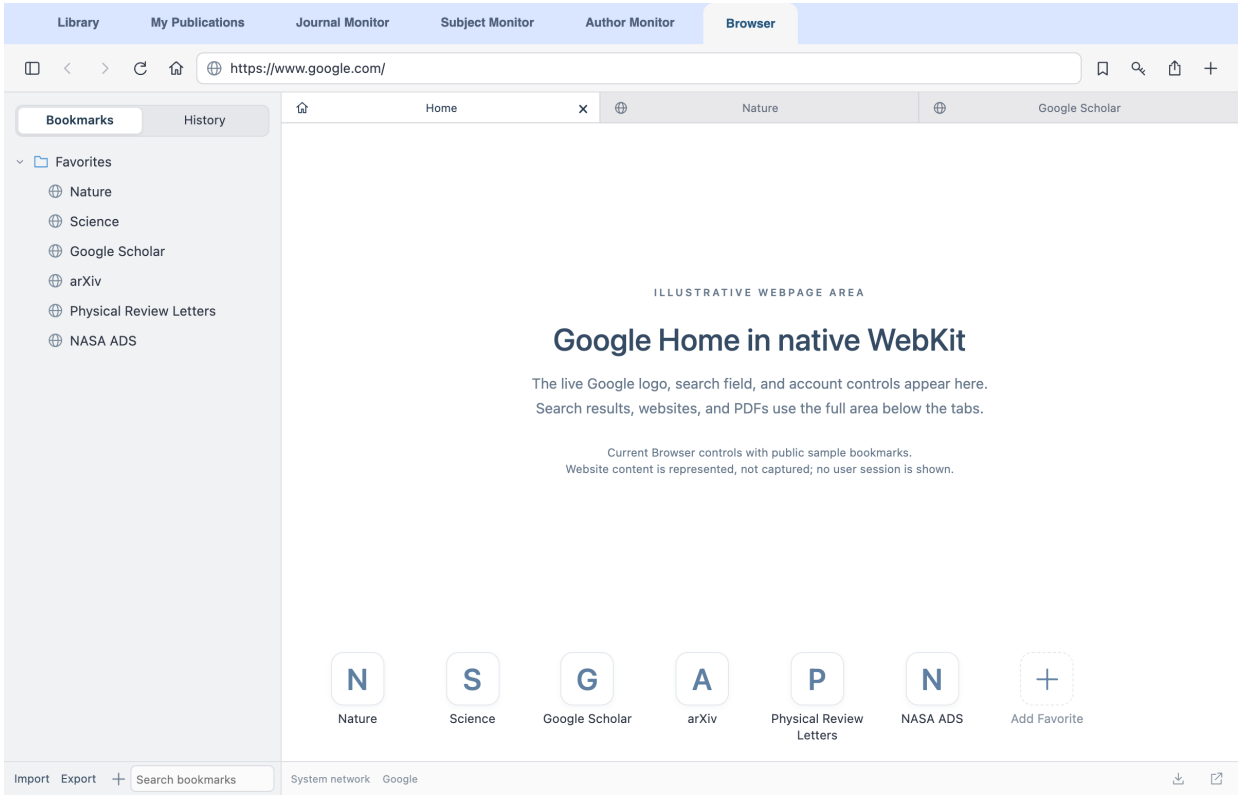
> **Note**
> Google Scholar requests are paced and protected by a cooldown. Repeated manual refreshes during a block do not improve the result.

See also: My Publications · Publisher and Network Access · Local Data and Privacy

3-8 Browser

3-8.1 Browse, Bookmark, and Revisit Websites

Browser is the sixth main tab, after Author Monitor. Website links across Library, My Publications and all three Monitors switch here, opening a new webpage tab without replacing existing pages.



Browser layout example. Rendered from the current Browser interface code with public sample bookmarks. The native webpage area is a labeled illustration, not a live Google screenshot. Website controls and favicons depend on the loaded site; no private session or history is shown.

Search and navigate

1. **Open Browser.** On first use, the main tab opens Home: real Google above local Favorites, using Browser's own session, not Safari's. Later switches return to your selected webpage; use the Home button to return to Google. A paper link opens its destination directly in the unused first tab. Nothing is preloaded during App startup.
2. **Use the address field.** Enter a website address or search terms. Plain search terms go to Google; an address opens directly. Use ⌘L to focus the field.
3. **Navigate normally.** Back, Forward, Reload/Stop, and Home act on the selected webpage tab. Favorites appears only on Google Home; search results, sign-in pages, websites, and PDFs regain the full content height.
4. **Switch between workspaces.** Returning from Library or a Monitor preserves the current page and browsing position without reloading merely because the main tab changed.

Return to the source workspace: ⌘[. After a website link in Library, My Publications, Journal Monitor, Subject Monitor, or Author Monitor brings you here, press Command-left bracket to return to that workspace's retained selection and scroll position. It works from the webpage or address field and leaves Browser tabs open; it is not webpage Back. The source lasts for this Browser detour only: manually leaving Browser clears it, and entering Browser directly does not create one. Close any open dialog first.

Instant Home, online search. Each tab lazily retains Home and a browsing view. Home returns without reloading; Reload refreshes it, including account state. App restart, interrupted loads and process

termination need a connection. Search/sign-in remain online. Tabs share Browser's persistent profile, without saved HTML/password copies or replayed forms.

When you search or follow a link from Home, Home remains visible while the destination begins loading, then gives way to that destination. It should not flash the page you visited before Home. A cancelled or failed pending navigation leaves Home available rather than presenting that older page as the result.

⌘F opens Find on Page. Enter moves to the next match; Shift-Enter moves to the previous match. Browser is available in the native **Gewu Insight.app**; opening the local UI in a separate web browser does not embed another browser engine.

Favorites on Home. Favorites/Favourites (including localized names) supplies the tiles; use **Create Favorites Folder** if missing. Open nested folders and return through breadcrumbs; use **Add Favorite** or right-click to edit. Google's real logo/search/account controls remain; secondary language/region/footer links are hidden only on Home. Native zoom fits Google above the divider-free tiles. Other pages keep normal zoom/scrolling. Excess tiles use Previous/Next, not vertical scrolling. Bottom bars align; hiding the sidebar keeps Favorites visible. Bookmarks are not sent to Google.

Use multiple webpage tabs

Below the address toolbar, adjoining rectangular tabs show icons, titles and × above the webpage column only. The sidebar begins directly below the toolbar; it is not pushed down by the webpage tabs. The top-right + (**New Tab**, ⌘T) opens Home. Switching preserves each page, position and Back/Forward history without reloading or stopping background loads.

Right-click links for **Open Link in New Tab**, or Command-click. New-window links also create tabs, including inline PDFs. Bookmarks/Favorites/History offer **Open in New Tab**. New tabs are selected immediately; ordinary clicks use the current tab.

× or ⌘W closes a tab; closing the last opens Home. Left/Right switches focused tabs. Up to 32 tabs stay in memory, not restored after App quit. Save drafts before closing; bookmarks, history, downloads and sign-ins remain. AutoFill only triggers in the visible page.

Save and organize bookmarks

The sidebar's top tabs switch between **Bookmarks** and **History**, without a repeated heading. **Import**, **Export**, + (New Folder), and Search sit along the bottom; History shows Clear History and Search there instead. Drag the sidebar's right edge to resize it or use the toolbar to collapse it. Width, tab, and expanded folders are saved locally.

1. **Add a page.** Use the bookmark toolbar button, **Share > Add Bookmark**, or ⌘D. The native sheet starts with the page title and address.
2. **Choose a location.** Edit the name or address and choose a folder under **Add to**. The previous destination is remembered. Saving the same address in the same folder reuses the existing bookmark.
3. **Create nested folders.** Use **New Folder**, or right-click a folder to create a child folder. The editor can also rename a folder or choose another parent.
4. **Arrange the tree.** Drag an item into the middle of a folder row to move it inside, or to a row edge to reorder it. Drop on the top **Bookmarks** tab to move to the root. A folder cannot move inside itself or its descendants.
5. **Save by dragging.** Drag the website icon beside the address field, or a webpage link, into a bookmark folder. Confirm the name and destination in the same native sheet.

Search filters bookmark names and addresses. Right-click a bookmark for **Open**, **Edit Bookmark**, or **Delete**. Deleting a folder requires confirmation and removes its contained bookmarks, not websites, browsing history, downloaded files, or papers. A bookmark is a saved link, not an offline webpage copy.

Import and export bookmarks

Choose **Import** at the bottom of Bookmarks and select a Safari or Chrome bookmark HTML file. Safari versions with **File > Export Bookmarks** produce HTML; newer Safari versions may use **File > Export Browsing Data to File** and produce ZIP. Select bookmarks only when exporting. Gewu accepts a ZIP containing exactly one bookmark HTML file and ignores all other members. In Chrome, use **Bookmark Manager > menu > Export Bookmarks**.

Imports go into **Imported Bookmarks**, retaining folders, order, names, and supported links. Repeating an import merges matching folder names and reuses the same URL in the same folder; existing bookmarks are not overwritten. The result reports added bookmarks, new folders, duplicates, and skipped links. Local addresses, application links, and recognized authorization links are skipped. Limits are 10 MB, 20,000 source entries, and 64 folder levels; invalid files do not partially import.

Export saves the full tree as standard bookmark HTML; right-click a folder for **Export Folder**. Safari and Chrome can import this format. Exports contain no history, cookies, or passwords, but their bookmark names and URLs are readable by anyone with the file. Keep personal exports private. Import/export is on demand, not ongoing Safari or Chrome synchronization.

Website icons

Bookmarks, Favorites and webpage tabs use a disposable local icon cache, separate from SQLite. Icons come directly from sites (at most two concurrent requests), without cookies, credentials, whole-article fetching or third-party icon services. Visits may supply a nonstandard icon URL. Missing icons use globes or Home initials; bookmark lists are never sent to an icon service.

Revisit and clear history

History groups loaded main pages by Today, Yesterday, and date, with names, sites, visit times, search, and pagination. Subresources, main-tab switches, and cached Home returns add no visits; Reload updates the existing record. Click to revisit; right-click to bookmark or delete a visit.

Clear History confirms Last hour, Last 24 hours, Last 7 days, or All history. It preserves bookmarks, downloads, cookies, and sign-ins. History remains until deleted; authorization callbacks and recognized signed-download URLs are excluded.

Address suggestions from your history

As you type in the toolbar address field, Gewu searches saved local visit titles and URLs. Address prefixes rank first, followed by matching titles; visit frequency and recency help order the results. A matching address can complete the typed prefix with its remaining text selected. These are local history suggestions, not live Google suggestions, and typing does not send a suggestion request to a search engine.

Use ↑ / ↓, then Enter to open a result, or click it to open directly. Escape dismisses suggestions and restores your typed query. Without a chosen completion, Enter opens an address or searches Google normally. Clearing history also removes those visits from suggestions; bookmark search remains a separate sidebar control.

Read PDFs and save downloads

Viewable PDFs stay inline, including attachment/download links, with article/PDF Back/Forward navigation. Save via Download or **Share > Save Download**; unsupported attachments use a save sheet. This uses Browser's session, not Python's downloader. For HTML viewers, use the site's PDF link. Downloads create no Library paper or Gewu Key: import explicitly. Library PDFs still open in macOS Preview.

System networking and Clash rules

WebKit follows macOS networking without changing proxies/Clash. When system proxy or TUN routes traffic to Clash, its rules choose DIRECT/proxy. **System network** is policy, not proof of routing. DIRECT needs a school-recognized exit or institutional login. PDF/login/CDN domains match separately; browser-specific extensions/process rules are not inherited. Verify Clash connections; see Publisher and Network Access.

Privacy and compatibility boundaries

Bookmarks/history use local SQLite; cookies/cache use Browser's persistent profile, not Safari synchronization. Browsing/search reaches websites, but external pages have no Library/credential bridge. Local/file/application addresses are blocked. HTTP recovery retries once as same-host HTTPS GET, without replaying forms or bypassing certificates; errors include reason/code. The local server rejects cross-site API requests and unrecognized hosts.

Use macOS Passwords

Open Passwords opens Apple's app (or use Passwords in System Settings on older Macs). Unlock and choose credentials there yourself. Gewu has no password vault and never imports, exports, lists or relays credentials.

Conservative AutoFill trigger. A single, unmodified click on an explicitly marked HTTPS login username or current-password input can open **AutoFill > Passwords**. Search, ordinary email, registration, codes, embedded frames and uncertain fields are skipped, as are keyboard focus, double-clicks and dragging. Each field kind is offered once per site per App session, including cancellation; right-click AutoFill retries manually. Navigation or intervening input cancels a pending trigger. macOS handles authentication, selection and filling; Gewu never reads credentials or submits login. Site matching, Safari-style suggestions and successful filling are not guaranteed.

A macOS prompt for **Gewu Insight WebCrypto Master Key** concerns WebKit's protection of locally stored webpage cryptographic keys, not a metadata API key. Local development rebuilds can require renewed keychain permission. Handle this system prompt yourself; do not share your Mac password or delete the key to suppress the dialog.

Password versus passkey. A Password entry contains a text password; a Passkey entry is a different sign-in credential and may show only a creation date. A passkey cannot be inserted as text into a password field. Use the website's passkey sign-in flow where supported, or select a genuine Password entry yourself. See AutoFill troubleshooting; the trigger is not a guarantee that macOS or a particular website can fill the field.

> **Website links**
> Publisher, Google Scholar, author/profile, preprint, citation-source, Note and remote-PDF links open here, as does Overleaf export. Local PDF/Preview and Finder actions stay native. For incompatible sites, explicitly use **Open in Safari**; cookies are not transferred. The browser-hosted UI opens ordinary tabs. Never bypass access controls.

3-9 Settings and Safety

3-9.1 Open Source Settings

Built-in sources work without setup. Optional credentials improve access or coverage for supported services.

Open settings

Choose **Setting > Open Source Settings**, or press ⌘,.

Available services

Service	Configuration
Crossref, arXiv, DataCite	Built in; no key field is required.
OpenAlex	Optional OpenAlex API key.
Semantic Scholar	Optional API key. Anonymous access remains available.
NASA ADS	Optional API token for astronomy and physics coverage.

Save a credential

1. **Paste the credential into its service field.**
2. **Choose Save beside that service.** Each service has its own Save, Test, and Clear controls.
3. **Choose Test and read the status underneath.** This checks service access; it does not refresh every paper. Then refresh one paper to inspect the available citation evidence.

A configured field shows fixed round dots by default, while the detailed status below continues to report saving, connection, or quota information. The dots do not expose the key or its length. Click **Show** inside the field to reveal the saved key, or inspect a newly typed key; click **Hide** to mask it again. A saved key is retrieved only on Show, and its displayed value is cleared on Hide or when settings closes. Type a replacement and choose **Save** to change the key; **Clear** removes the locally saved key. A configured environment key can remain available. Avoid screenshots or screen sharing while a key is visible.

Semantic Scholar sends its key in the official `x-api-key` header. Requests share a low-rate queue and use exponential backoff after rate limits or temporary failures.

OpenAlex status reports the remaining daily free allowance and its reset time when available. Gewu uses only that free allowance and falls back to other metadata sources when it is exhausted; it does not spend paid OpenAlex credits. A saved key is not proof that every remote request will succeed.

> **Important**
> Credentials are local-only: store them in local SQLite or approved environment variables. Do not place keys in source files, logs, exported reports, tests, or Git.

See also: arXiv Scope Settings · Field Settings · Local Data and Privacy · Publisher and Network Access

3-9.2 arXiv Scope Settings

Choose one or more arXiv subject archives for discovery. Selections can span several research fields or combine several groups within one field.

Open arXiv Scope Settings

Choose **Setting > arXiv Scope Settings**. The window lists every supported primary subject and its archive groups. Select at least one group, then choose **Save**.

Understand the hierarchy

The first level identifies a broad arXiv subject such as Physics, Computer Science, Mathematics, Economics, Electrical Engineering and Systems Science, Quantitative Biology, Quantitative Finance, or Statistics. Each checkbox represents an archive group beneath that subject. For example, **Physics > Condensed Matter** includes all `cond-mat.*` leaf categories.

You may select several groups under Physics, or combine groups across primary subjects. Categories shared by more than one scope are fetched only once.

Default and saved behavior

- The default selection is only **Physics > Condensed Matter**.
- **Restore Defaults** changes the selection in the window; choose **Save** to apply it.
- Saving changes future arXiv refresh coverage and rebuilds the available **Preprint Filter** categories.
- Previously saved papers remain in SQLite. Changing scopes never deletes earlier discoveries.

> **Note**
> The category checkboxes in Journal Monitor are display filters. They do not narrow the discovery scope selected here.

See also: Journal Monitor · Field Settings · Open Source Settings · Publisher and Network Access

3-9.3 Field Settings

Gewu Insight keeps a detailed canonical field classification while letting each workspace decide which fields deserve their own visible filter.

Open Field Settings

Choose **Setting > Field Settings**. Select a field to show it as a separate filter and label. Clear it to collect that field under **Other Fields**.

Default organization

Shown separately by default	Grouped under Other Fields by default
Physics	Astronomy & Astrophysics
Quantum Information	Particle & Nuclear Physics
Materials Science	Mathematics
Chemistry	Computer Science
	Biology & Medicine
	Statistics
	Economics & Finance
	Electrical Engineering
	Earth & Environmental Science
	Social Sciences
	Unclassified records

How the visible filters behave

- **Other Fields** appears immediately above **All Fields**.
- **All Fields** is always the final choice and includes every classified paper.
- Changing this setting updates field filters, counts, badges, and Paper Details across Journal Monitor, Subject Monitor, Library, My Publications, and Author Monitor.
- The setting changes presentation, not evidence. Canonical field assignments remain in SQLite and become visible again if a field is later selected.

> **Note**
> Use **Restore Defaults** to return to the four-field default. Field Settings do not change arXiv discovery coverage; configure that separately in **arXiv Scope Settings**.

See also: arXiv Scope Settings · Open Source Settings · Journal Monitor · Subject Monitor

3-9.4 Quick Look Settings

Choose whether Space-bar previews stay inside Gewu Insight, use macOS Quick Look, or divide the work by file format.

Open Quick Look Settings

Choose **Setting > Quick Look Settings**. The setting applies to Space-bar previews in both Library and Spaces. If a preview is already open, saving a different mode refreshes that preview with the newly selected behavior.

Mode	Behavior
Gewu Insight for all files	The default. Every preview stays inside the workspace. PDFs use the built-in reader; unsupported formats keep the built-in file card and **Open File** action.
Gewu Insight for PDFs, macOS for other files	PDFs use the built-in reader. Presentations, media, images, and other formats use the macOS Quick Look panel.
macOS Quick Look for all files	PDFs and every other format supported by macOS use the system Quick Look panel.

Choose a mode for your workflow

- Use the default when you want one consistent, contained workspace and the optimized built-in PDF reader.
- Use the mixed mode when PDFs are your main reading material but presentations, video, audio, and images should use native macOS preview support.
- Use the all-native mode when system Quick Look compatibility matters more than keeping the preview embedded in Gewu Insight.

> **Note**
> When the macOS panel is used, keyboard focus remains in Gewu Insight so ↑ and ↓ continue changing the selected Library or Paper Files item instead of turning into page-navigation commands in the system panel.

See also: Preview Files with Quick Look · Understand Paper Files · Navigate with Mouse and Keyboard

3-9.5 Naming and PDF Rename Rules

Display naming and physical PDF renaming are separate choices. Review metadata before applying either one in bulk.

Choose the Library naming rule

Choose **Setting > Naming Rule > Use Title** or **Use Citation**. This changes the preferred managed display/name rule where applicable.

Configure PDF renaming

Open **Setting > PDF Rename Settings**. A template can use tokens such as:

- `{author}` or `{authors}`
- `{year}`
- `{title}`
- `{journal}`
- `{doi}`

Rename one item from metadata

1. **Confirm its DOI, title, author, journal, and year.**
2. **Control-click the item.**
3. **Choose Auto rename from Metadata.**
4. **Verify the result in the Library and Finder.**

> **Note**
> Changing a Paper Files property or choosing **Set as Default** does not rename that file. A rename occurs only through a naming or rename action.

See also: Choose File Types and the Default File · Folders, Shortcuts, and Spaces

3-9.6 How Gewu Insight Stores Files

The visible Library stays simple: its main tree contains a compatibility primary path. Other resources can live in a mirrored Associated Files area, while the UI presents every resource as a peer in one flat Paper Files list.

One work, several representations

Each managed paper has one database identity and a stable **Gewu Key**. DOI and arXiv are external identifiers attached to that identity; neither is used as the sole storage identity.

Where the files go

- A compatibility primary PDF stays in the visible Library, Space, or My Publications tree.
- Older PDF versions go to a mirrored `Versions` area.
- Supporting information goes to `Supplements`.
- Images, notes, presentations, data, code, and other attachments go to `Files`.

The associated bundle mirrors the paper's managed path under data/AssociatedFiles. When you move the paper in the UI, Gewu Insight moves the associated bundle and refreshes live paths.

Why the UI and Finder look different

The UI presents a paper as one item with a flat Paper Files list inside it. Finder shows a compatibility primary file in the normal tree and the related bundle in Associated Files. The UI Default marker is a reading preference and does not require physical file promotion or renaming.

Copies and filesystem shortcuts

When the same work appears in more than one Library folder, those locations can share one database record and one canonical managed file. Symbolic links can represent secondary filesystem locations. **Make Copy Desktop** resolves those shortcuts into real files for a portable export.

> **Important**
> Do not manually move an Associated Files bundle or replace an alias in Finder. Use Gewu Insight so the database and physical paths change together.

See also: Storage Reference · Make a Portable Desktop Copy · Understand Paper Files

3-9.7 Local Data and Privacy

Gewu Insight runs a local interface and keeps its authoritative library database on your Mac. Network requests occur only for the features that need external metadata, pages, profiles, or files.

Local components

- The UI server listens on the local loopback address, normally `127.0.0.1:8765`.
- SQLite stores papers, identities, relationships, settings, and operational state.
- Browser bookmarks, folders, history, and sidebar preferences also live in SQLite. Browser cookies and cache use a separate persistent WebKit profile; they are not shared with Safari or exported with the database.
- Managed PDFs, attachments, photos, and generated reports stay in local project directories.
- JSON files are exports or diagnostics; they are not the authoritative database.

External requests

Metadata refresh, citation counts, journal/subject/author discovery, profile images, website opening, and downloads can contact their named services. The app paces services that require strict request limits.

Protect credentials

Optional service credentials belong only in Open Source Settings or approved environment variables. They must not appear in a screenshot, support message, Git commit, or exported diagnostic file.

See also: Open Source Settings · Back Up or Move to Another Mac · Publisher and Network Access

Part IV.

Product Identity

4-1 Visual Identity and Cultural Elements

Gewu Insight uses three related marks—the calligraphic Chinese name, the app icon, and a seal-script watermark—to connect a contemporary research workspace with the longer Chinese tradition of examining things in order to form insight.

The marks do not decorate every surface. Each has a different job: the icon identifies the application, the calligraphic name gives the product a human signature, and the quiet seal-script pattern belongs to transitional or otherwise empty space.

4-2 The calligraphic name



The transparent production artwork used in the About panel. From left to right the picture contains 見、洞、物、格; read in the traditional direction from right to left, it says 格物洞見.

This is a **祝枝山行草集字** composition: four separately documented historical character forms were assembled into one wordmark. It is therefore accurate to call it "a collection of characters in the calligraphic style of Zhu Zhishan," but not "a title written by Zhu Zhishan." No surviving single inscription titled 格物洞見 is being claimed.

4-2.1 How the four characters were assembled

Reading order	Reference character	Recorded hand and script	Role in the product artwork
1	格	祝枝山 · 行书	Begins the product name when read from the right.
2	物	祝枝山 · 草书	Completes the classical term 格物.
3	洞	祝枝山 · 草书	Introduces 洞見, insight gained by seeing through a subject.
4	見	祝枝山 · 草书	Closes the traditional form of the name.

The selected glyphs were optically balanced into a common line: scale, spacing, ink tone, edge cleanup, and transparent alpha were normalized while the characteristic strokes were retained. The production preparation used a Codex-assisted image workflow. The interface and accessibility text use the contemporary simplified product name **格物洞见**; the artwork deliberately preserves the traditional character **見** and traditional right-to-left composition.

> **Provenance and reuse**
> The character links above are retained as provenance for the design process. The source thumbnails do not state a clear open-reuse licence. They should not be treated as public-domain assets merely because they are viewable online. The assembled Gewu Insight product artwork is distributed under the repository's all-rights-reserved notice.

4-3 The app icon and the seal-script watermark

4-3.1 The app icon



The icon's two blue-violet fields read simultaneously as an open book and two sides of a field of view. The white eye and central point suggest observation and focus; the tapered center also recalls a pen nib or the spine of a book.

This description is the product's visual reading of the finished mark, not a claim that every interpretation was specified in an original design brief.

4-3.2 The seal-script watermark



The 2×2 motif is a digital rendering in 汉仪篆书繁. Read down the right column and then down the left, its four characters are 格、物、洞、见. It uses simplified 见 because that is the supported glyph in the selected digital typeface.

The mark is a light-line screen motif inspired by seal composition, not a photograph or facsimile of a historical carved seal. Typeface licensing remains separate from the product artwork.

4-4 Why the three marks are used differently

Mark	Role
Icon · recognition	Use at application scale: Dock, startup identity, About, and platform packaging.
Calligraphy · signature	Use where the Chinese name can be read deliberately, especially the About panel and identity materials.
Seal script · atmosphere	Use at very low opacity in startup transitions or quiet lower-right space, never where it competes with paper details.

Together the three registers establish the product's visual identity: the icon is contemporary and operational; the calligraphy is personal and gestural; the seal-script watermark is historical in reference but restrained in use. Their common purpose is to make the name visible without turning the research workspace into a themed illustration.

See also: Welcome · The Personal Research Workspace · Startup, Monitor Cache, and Text Size

Part V.

Technical and Operational Reference

5-1 Keyboard Shortcuts

Use standard macOS editing commands throughout the app. The shortcuts below are specific to navigation and Gewu Insight commands.

Shortcut	Action
↑ / ↓	Move through the focused Library or Paper Files list; on the source rail, move among Library, shortcuts, and Spaces and open the highlighted source.
→	Enter the highlighted source from the vertical rail, enter a folder, or move from a selected paper into Paper Files.
←	Leave Paper Files for the selected paper, return to a parent folder, or move from the leftmost Library column to the source rail.
Space	Open or close Quick Look for the selected local paper/Paper Files item, including available monitor PDFs; not while editing text.
Esc	Dismiss an active transient control or return focus where supported.
⇧⌘V	Toggle List View and Table View.
⌘L	Focus the address/search field in Browser; otherwise open Folder Loader.
⌘D	In Browser, bookmark the current page.
⌘R	In Browser, reload the current page.
⌘F	In Browser, Find on Page. In Library, My Publications, and the three Monitors, focus the current workspace search.
⌘T	In Browser, open a new webpage tab at Home.
⌘[	Return from Browser to the source workspace; keep its position and webpage tabs.
⌘,	Show Open Source Settings.
⌘W	In Browser, close the selected webpage tab; closing the last creates Home. Outside Browser, dismiss active Quick Look first, otherwise close the current window.
⌘Q	Quit Gewu Insight. Save webpage drafts first; open Browser tabs are not restored after quitting.

5-1.1 Monitor keyboard navigation

In a Journal or Subject calendar, Left/Right moves one day and Up/Down moves one week. Enter selects the date and enters the papers. In the paper area, Up/Down selects papers, Left/Right changes the journal or subject, and Escape returns to the calendar. Space opens an available local PDF in Quick Look.

In Author Monitor, Left selects the author rail and Right selects its papers; Up/Down moves through the active pane. Up from the first item enters the main-tab navigation level. At that level, Left/Right chooses a main tab and Down or Enter returns to its content. These commands do not apply while typing in an input.

5-1.2 Browser keyboard context

Find on Page uses Enter for the next match, Shift-Enter for the previous match, and Escape to close. In address suggestions, Up/Down selects a result and Escape restores your query. A focused webpage tab uses Left/Right to change tabs. These contexts are separate from Library navigation; ⌘L in Browser does not open Folder Loader.

5-1.3 Paper Files mouse behavior

Single-click selects. Double-click opens. Control-click opens the Paper Files context menu. This is intentionally different from a one-click launcher.

See also: Navigate with Mouse and Keyboard · Understand Paper Files

5-2 Publisher and Network Access

Publishers can rate-limit shared VPN addresses or automated-looking traffic. Gewu Insight reduces overlap and request rate, but it cannot remove a publisher block.

5-2.1 If a publisher page is blocked

1. **Stop repeated refreshes.** Additional requests can extend a temporary block.
2. **Try your direct connection.** If the page works with VPN off, the VPN exit IP is likely the blocked client.
3. **Use Open Website or an external browser.** Complete any normal institutional login or publisher interaction yourself.
4. **Download only when authorized.** Add a legitimately obtained PDF with **Add File**.

5-2.2 Why figures may arrive later

Figure and article-page retrieval is queued behind discovery work and uses caching. This reduces simultaneous RSS, page, and image requests to the same publisher. A paper can appear before its representative image.

5-2.3 Browser and your system network

The Browser tab uses native WebKit requests and does not proxy pages through the Python server. System proxy or TUN capture can send its traffic into Clash; only then do the active rules determine DIRECT or proxy routing. A DIRECT rule uses your current physical network, which must itself provide institutional access. Gewu Insight neither changes these rules nor guarantees a particular route; use Clash connection records to verify the actual route. See Browse, Bookmark, and Revisit Websites for downloads, sign-in limits, and the separate browser profile.

5-2.4 Service rate limits

Semantic Scholar, Google Scholar, and some publisher sources use shared pacing, cooldowns, or exponential backoff. Let the bottom status area settle before starting the same action again.

> **Important**
> Do not attempt to bypass a paywall, CAPTCHA, browser verification, or access control. Use your authorized browser session or attach a file you obtained legitimately.

See also: Journal Monitor · Download and Sync Versions · Troubleshooting

5-3 Startup, Monitor Cache, and Text Size

The macOS app starts a local UI server, waits for the Library bootstrap, and then opens the main window.

5-3.1 Normal startup

The startup panel remains visible while the local service becomes ready and the first Library view is prepared. It may show more than one status message, but the panel should remain visually stable.

The first launch after an additive database upgrade can take longer because the existing SQLite records are migrated and older provenance or file metadata is backfilled. The native launcher allows up to 120 seconds for this preparation. Keep the single startup panel open rather than launching a second copy; the migration preserves the existing database and managed files.

The Dock tile, native startup panel, and in-window loading state use the icon packaged with the same app build. Startup has a native launch phase and a local-web bootstrap phase, but the visual handoff should not introduce a second icon frame or fall back to an older product mark.

5-3.2 Automatic local-server recovery

If the local UI server exits unexpectedly while the macOS window remains open, the launcher starts it again. The window rechecks local health and reloads the workspace when the service becomes ready. A startup-failure sheet is nonblocking; if the server recovers after that sheet appears, the stale sheet is dismissed automatically instead of continuing to interrupt a healthy workspace.

A brief reload during recovery does not mean the SQLite database was rebuilt or that another copy of Gewu Insight should be opened. If repeated recovery never reaches a healthy Library view, use the launcher log steps below.

Recovery reloads the workspace, so Browser's memory-only tabs and unsubmitted forms can be lost. Bookmarks and history already saved in SQLite remain. Save work on websites regularly; recovery is not a web-session or draft backup.

5-3.3 When automatic updates run

Automatic monitoring runs inside the local UI service, with startup delays, periodic checks, and saved completion/retry state. Keep the Mac awake and the service running. Closing a window does not necessarily stop that service. Quitting the App stops a server it started; a separately launched server must be stopped separately. A late launch checks work that is due rather than guaranteeing a run at an exact minute.

Monitor	Current automatic schedule
Journal Monitor	Daily work becomes due from 09:00 in the Mac's local time zone; stored completion and pending retry state determine what still needs work.
Subject Monitor	Daily work becomes due from 09:00 local time; incomplete sources have their own bounded retry state.
Author Monitor	Checks due authors from 10:00 local time; already checked authors are not blindly fetched again on every scheduler tick.
arXiv	Uses its separate announcement-ready schedule and buffer, not the journal's 09:00 rule. Read the displayed update/countdown state.

These are local scheduler rules, not guarantees of remote source completion or a hosted monitoring service. Manual Refresh remains available for the selected scope; repeated broad refreshes can worsen a source's rate limit.

5-3.4 If the app cannot start

1. **Quit any old Gewu Insight window.** An earlier local server may still own port 8765.
2. **Open the launcher log.** Check logs/paper_monitor_launcher.log for the first error.
3. **Allow the first startup to finish.** After an upgrade, wait through the migration window before treating the preparation panel as a failure.
4. **Start the app again once if recovery does not complete.** Avoid opening several copies while the first is still preparing.
5. **Preserve the database.** Do not delete SQLite as a startup fix.

5-3.5 Clear a stale interface

Choose **Setting > Clear Monitor Cache** if a monitor still shows stale downloaded images or rendered web assets after an app update. This clears disposable monitor presentation cache, not the SQLite database, saved paper records, or managed PDFs.

This is not Browser's Clear History command and does not clear Browser sign-in cookies or Apple Passwords. Use **Browser > History > Clear History** for visit records only.

Monitor cache and macOS app identity are separate. Clearing Monitor Cache cannot repair a stale Dock tile. After replacing the app bundle, quit every running copy and open the rebuilt **Gewu Insight.app**. If a permanently pinned Dock tile still points to an older bundle at another path, remove that tile and add the current app again.

5-3.6 Choose a comfortable text size

Choose **Setting > Font Size**, then select **Compact**, **Standard**, **Large**, **Larger**, or **Extra Large**. The workspace previews the choice immediately while keeping panel dimensions, icons, and window controls unchanged.

Cancel, the Escape key, or clicking outside the sheet restores the previously saved size. Choose **Save** to keep the preference in the local SQLite settings. New app sessions apply the saved size automatically.

See also: Troubleshooting · Local Data and Privacy

5-4 Troubleshooting

Start with the visible status and the selected work. Avoid broad repair actions until you know whether the problem is network, metadata, database state, or a local path.

5-4.1 A PDF cannot be found after a move

1. Reselect the paper and try **Show in Finder**.
2. If the move happened in Gewu Insight, close and reopen the item to refresh the view.
3. If the file was moved manually in Finder, return it if possible, then organize it through the app.
4. Check Recycle before importing another copy.

5-4.2 Paper Files shows Missing File

This label is a path-state warning, not a failed metadata lookup: SQLite still knows the resource, but the recorded file path does not currently exist. Check **Show in Finder**, the corresponding data/AssociatedFiles bundle, and Recycle. If the file was moved outside Gewu Insight, restore it before reorganizing it through the app. Do not delete the row merely because its filename resembles the default PDF; explicitly typed attachments and the default resource remain visible for recovery.

5-4.3 Quick Look is slow on the first open

A large or previously unseen PDF may need an initial structural read and thumbnail render. The main document should appear before the full thumbnail strip. Leave the preview open briefly instead of pressing Space repeatedly; later opens can reuse page information and cached thumbnails. If the viewer never produces a page, use **Open in Preview** to check whether the file itself is valid.

5-4.4 The wrong file is the default

1. Open Paper Files and verify every file type.
2. Correct a mislabeled Corrected published version, Published version, Accepted manuscript, or Preprint; expand the size footer to set annotation separately.
3. Choose **Set as Default** for the desired valid file, or choose **Use Automatic Selection** to resume automatic selection.

5-4.5 A duplicate appears after download

Compare file size and classification. New downloads use exact SHA-256 and PDF content fingerprints. A copy with annotations may be retained intentionally. Exact byte duplicates are reused, while clean and annotated peers can both remain. Keep the intended default before deleting any redundant non-default row.

5-4.6 A monitor shows no papers

Read the source status. An empty successful source is not the same as a failed fetch. In Subject Monitor, allow scheduled source-level retries to finish, or use the selected subject's refresh command when a manual retry is needed. In Journal Monitor, confirm the date and filters. If a VPN is involved, test a direct connection without repeatedly refreshing.

5-4.7 A journal count and the visible cards disagree

This is usually expected when **Field Filter** is active. Journal counts describe all stored papers for the selected journal and day; field choices narrow the visible cards. Clear or broaden the field selection before diagnosing a missing fetch.

5-4.8 A journal disappeared from the configured list

Open **Journal Settings** and inspect the disabled history before editing configuration files. Removing a journal disables its subscription but preserves previously collected papers and history. Add it again to restore monitoring. Because the subscription list is SQLite state, replacing source files should not be used as a journal-list repair.

5-4.9 LaTeX remains visible as source text

On a cold launch, allow the math renderer a moment to become ready. The active day should re-render automatically. If raw `$...$` text remains, check network availability for the renderer, then reload the same view once; switching calendar dates should not be required.

5-4.10 Folder Loader looks incomplete

Open the live or previous Loader log and the result guide. Confirm the metadata mode, source path, and any stop action. Re-running the same source should reuse existing files rather than overwrite the primary PDF.

5-4.11 A default file cannot be deleted

The preferred Default and the protected physical primary file are not always the same resource. **Set as Default** changes the reading preference, not the Library's compatibility primary path; switching defaults does not guarantee that the old file becomes deletable. Remove ordinary non-primary resources through Paper Files. To remove the managed paper itself, use the normal Library reference/Recycle workflow, which preserves shared references. Do not bypass a primary-file warning in Finder. An **Invalid file** remains an audit record and is never eligible to become default.

5-4.12 A website link opens Browser instead of Safari

This is now the normal action for web links in all five research views, including publisher arrows, DOI/arXiv links, scholarly profiles, citation-source links, and links inside Note. Browser selects a new webpage tab, or reuses its unused first tab. Existing web pages stay open. Local PDF, Preview, Finder, download, and file-import actions are unchanged. Use Browser's explicit **Open in Safari** only when you want the external browser.

5-4.13 Browser Home needs a connection or a website will not load

A warm Home returns instantly from memory; it is not an offline copy of Google. First use after quitting, an explicit Reload, or a terminated WebKit process needs a connection. Search results and account actions always require the website. Read the actual load error, check the destination and Clash connection route, and try Reload once. A website may reject embedded browsers or have its own session policy; an unsupported-browser banner alone does not prove why a sign-in ended. Open in Safari is a manual compatibility fallback, with a separate sign-in session.

5-4.14 AutoFill does not appear or does not fill

Automatic triggering is deliberately limited to an explicit click in a clearly marked login field. Ordinary email, search, registration, verification-code, and uncertain fields do not qualify. Cancellation uses up the automatic offer for that field kind/site during the App session. Right-click the actual input and choose **AutoFill > Passwords** to retry manually.

Unlock and select the appropriate entry yourself. If the item says **Passkey** and shows a creation date, it is not a text password. If a real Password entry closes without filling, macOS or that website may not support the return path in this embedded view. Gewu does not read your vault, relay credentials, or submit forms to work around it. Use the website's supported login flow or Safari; do not send passwords or unmasked credential screenshots in a support report.

> **Important**
> Before any manual filesystem repair, quit the app and make a full backup of the database and managed files together.

See also: Back Up or Move to Another Mac · Startup, Monitor Cache, and Text Size · Publisher and Network Access

5-5 Storage Reference

These paths describe the managed project layout. They are useful for backup and diagnosis, not for routine manual reorganization.

Location	Purpose
data/paper_monitor.sqlite	Authoritative database for papers, identities, relationships, settings, operational state, Browser bookmarks, and visit history. May contain saved metadata API keys; treat backups as private.
data/cache/journal_logos	Rebuildable local image cache for configured Journal Monitor logos.
data/Library	Visible main Library root and compatibility primary files.
data/Spaces	Independent Space roots.
data/MyPublications	Managed local files for My Publications.
data/AssociatedFiles	Mirrored bundles for versions, supplements, and attachments.
data/pdfs	Legacy managed PDF location retained for compatibility where needed.
`reports`	Generated reading notes and reports. These are artifacts, not the database.
logs/paper_monitor_launcher.log	macOS launcher diagnostics when the local UI cannot start.

Quick Look page thumbnails use a separate, automatically trimmed temporary cache. It is rebuildable preview data, not part of the Library, Associated Files, or the SQLite record, so it does not belong in a research backup.

Browser website data and its disposable favicon cache are managed by macOS outside the project. They are separate from SQLite bookmarks/history and from the monitor caches listed above. See Browser and credential backup boundaries before moving to another Mac.

5-5.1 Associated Files layout

An associated bundle follows the managed scope and path, then a work-specific bundle. Its children separate `Versions`, `Supplements`, and `Files`. The exact physical name can change as the paper moves; the Gewu Key remains stable.

5-5.2 Database and exports

SQLite is the only source of truth. JSON, HTML, Markdown, Word, LaTeX, and PDF outputs are exports, diagnostics, or reading artifacts. Editing one of them does not edit the Library record.

> **Important**
> Do not delete, rebuild, or "clean" the database because a generated JSON file looks stale. Diagnose the selected record and local path first.

See also: How Gewu Insight Stores Files · Back Up or Move to Another Mac · Local Data and Privacy

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